Trends in Ecology and Evolution

A systems-modelling approach to predict biological responses to extreme heat

--Manuscript Draft--

Manuscript Number:	TREE-D-25-00220R1
Article Type:	Opinion
Keywords:	plant-animal interactions; microclimate; biophysical models; thermal load sensitivity; dynamic energy budget models; coexistence theory; ecological modelling; population dynamic models
Corresponding Author:	Daniel Noble Australian National University Canberra, AUSTRALIA
First Author:	Daniel Noble
Order of Authors:	Daniel Noble
	Margaret Mayfield
	Ary Hoffmann
	Zhong-Hua Chen
	Steven J Lade
	Xuemei Bai
	Danielle Way
	Belinda Medlyn
	Owen Atkin
	Adrienne Nicotra
	James M. Cook
	Brajesh Singh
	Alison Bentley
	Ian Wright
	Michael R. Kearney
Abstract:	Anthropogenic climate change is leading to more frequent and extreme heat waves. These large-scale events are radically re-shaping interactions among organisms – impacting biodiversity, community composition and ecosystem services crucial to natural systems and food security. Predicting heat wave impacts on interacting species requires an understanding of the processes driving differential exposure and sensitivity of organisms to extreme heat events in a life-cycle context. To achieve this predictive capacity, we need to integrate models across scales while capturing species-specific responses at the individual level. We review and demonstrate how existing models in disparate fields can be linked to achieve an increased understanding of how individuals and communities will respond to extreme heat, now and into the future.

Australian National University

Dear Dr. Stephens,

We are pleased to submit our revised manuscript entitled: "*A systems-modelling approach to predict biological responses to extreme heat*" as a Review article in *Trends in Ecology and Evolution*.

We are excited about the uniqueness of our contribution. It reviews and articulates the innovative new theoretical developments across different fields and how they can be integrated to allow us to predict the impact of extreme climates on organisms, populations and communities. Our Review provides a critical overview of new modelling approaches across different fields that is unlikely to be found elsewhere. We believe such an integrative piece will be influential to how predictive models are developed in the future. Importantly, our paper brings in case studies to demonstrate how such a systems-modelling approach can be implemented – an important aspect in showing its power to capture feedbacks and making predictions relevant to whole systems.

We would like to thank you and the three reviewers for thorough reviews of our manuscript. We have substantially revised our manuscript to accommodate the many very constructive points raised. Our revisions include bringing in cases studies from the past supplement into Boxes 1 & 2 to highlight more clearly how a systems-modelling approach can be implemented, demonstrating the unique ways in which coupling and feedbacks can be built into a systems model, editing all figures and developing new ones to simplify and better capture all models discussed in the Review. We have also edited all text sections, removing text on models that have been discussed elsewhere along with re-writing our Boxes, Highlights and Concluding Remarks sections. We believe our manuscript is much stronger now and will make an impactful contribution to *Trends in Ecology and Evolution*.

In accordance with the Review guidelines, our revised manuscript has 5 display items (3 boxes and 2 figures), 109 citations and is 3,587 words. Some additional citations were necessary to capture some newer papers recently published and address comments from reviewers. We would like to thank you again for considering our manuscript.

All the best,
Dr. Daniel Noble

On behalf of all coauthors:

Professor Micheal Kearney
Professor Belinda Medlyn
Professor Ary Hoffmann
Professor Owen Atkin
Dr. Alison Bently
Associate Professor Zhong-Hua Chen
Distinguished Professor Xuemei Bai
Professor Adrienne Nicotra
Professor Margie Mayfield
Distinguished Professor Brajesh Singh
Professor Danielle Way
Professor James Cook
Dr. Steven Lade
Professor Ian Wright

Dear Dr. Stephens,

We would like to thank you and the reviewers for their very constructive comments on our manuscript. As indicated in our cover letter, we have made substantial revisions to our manuscript to address the comments, including revising all figures, highlights, outstanding questions, glossary, two boxes, and substantial editing of the text to make the ideas clearer and more concrete. We also focus our Review on models not already covered in detail in other reviews. We believe these revisions have vastly improved the paper and we thank you all for your time in reading and commenting.

Below we provide a point-by-point response to each comment and indicate what we have changed. Our responses are in blue and we provide **line numbers** (in bold) to the changes in the main, ***cleaned and formatted version of the manuscript***.

For clarity, we also provide a track-changed version indicating the changes made. Please note that this version is not formatted with the correct references but does highlight clearly the areas of the text that we have changed. We have also labelled each comment from yourself and the reviewers with a unique number and indicate in the track changed version where these changes are in comment boxes.

Once again, thank you for consideration of our manuscript and hope you now find it suitable for publication in *Trends in Ecology and Evolution*.

Sincerely,
Daniel Noble (on behalf of all authors)

Editor

C1: Referee 1 and 2 both point out that you discuss different models but give no advice as to how one might go about choosing a model, particularly important for practitioners - how your ideas might be applied. This would be good replacement for the current summary in the "Concluding Remarks" section (see my comments). Many of Referee 3's comments indicate that you have made statement that "x should be done" but haven't provided the necessary information as to how to do this.

Response: Thank you. This is a great idea. We have now re-written our Concluding Remarks section to be more forward thinking, highlighting challenges going forward and some advice on building models (**lines 287-292**). Figure 1 was meant to identify the relevant models that can be interconnected but we have expanded on model selection in various locations of each section (e.g., also **lines 88-90**).

We have now brought into our manuscript the key steps from the supplement to show how to translate microclimates, to temperatures of organisms and showing how systems can be coupled and behaviour integrated (**see new Box 1**). Organism temperatures can then be used in physiological models to estimate life cycles, survival in response to extreme heat and then be used to estimate demographically relevant parameters (intrinsic growth rate) to understand population growth and species coexistence (**see new Box 2**). Demonstrating these keys steps are something all reviewers wanted to see in the main text, and we agree wholeheartedly with them on this point.

C2: Referees 2 and 3 feels that you spend too much time on certain aspects which have been covered by recent reviews and, as such, skim over what is more interesting and novel about your approach. Again, this will free up space to discuss broader applications of your approach.

Response: We agree. We have now dramatically focused the first section, removing the GCM section and trimmed down the microclimate modelling (**lines 74-106**). This has resulted in a more balanced approach across the modelling sections.

C3: Referee 3 comments on many of the figures and states that they found them hard to understand.

Response: We have adjusted our figures and figure legends as best as we can to hopefully make them easier to understand. For example, we've simplified **Figure 1** by removing detail that wasn't necessary and simplified **Figure 2** dramatically along with its legend. It wasn't always clear from reviewer 3's comments what stood out as unclear specifically, but we focused on the general point and sought places where simplifications could be made and did so.

Editorally, I have a number of points I need you to address:

C4: (1) Many parts of the paper are over the various word counts - every section must be within 10% of the stated limits (except the Abstract, where 120 words is a hard limit). This has been categorised as an Opinion article and it appears you have used the guidelines for a Review article. The main text is 3810 words and needs to be reduced to no more than 2750. You currently have five elements (2 figures and 3 boxes), you must remove one (Box 3 is the obvious candidate). All Boxes need to be reduced to no more than 440 words and the main text figure captions to no more than 250. Box figure captions must be short (i.e. less than 100 words) - they are currently okay.

Response: Sorry. There was some confusion about the article type as it was originally approved as an Opinion but changed to a Review between the acceptance of the proposal and the submission. As clarified with the Editor, this should be considered a Review. We have checked that our submission word counts and element numbers follow the Review submission guidelines here: https://www.cell.com/trends/ecology-evolution/authors.

All our boxes are now < 420 words, with figure box captions all being less than 100 words. We also simplified figure legends in the main text, and the total work count for the body of the text is 3,587 words. We have slightly more than 100 references (109 references to be exact), but we thought this was important to address reviewer comments and also we included some recent papers published that were important.

C5: (2) The Concluding Remarks need rewriting. You have just provided a summary of the paper. The Concluding Remarks section should aim to tie threads together and indicate the path forward

Response: Thank you. Good suggestion. We have now re-written our concluding Remarks section to be more forward looking, discussing the challenges to a systems model approach, some solutions and ways to get started (in terms of model choice) (**lines 273-295 & 279-281**). Model choice is difficult to advise on specifically because it will depend on what processes a researcher and modeller deem important for a given system. As such, we have kept this advice a little more general and referred to other reviews on this topic.

C6: (3) The Supplementary Information is to go. I appreciate that you have put a lot of work into it but it is too extensive for a TREE article. You have essentially an additional paper in there. Either the information is important enough that it is brought into the text as a Box or it isn't important. Very few readers will access it (we have download figures).

Response: We have moved the crucial elements out of the supplement in the main manuscript now into two completely new boxes. More specifically, creating a new Box 1 and Box 2 which work together to demonstrate how a multi-species systems-modelling approach can be achieved. This was something Reviewer 1 and 2 liked and wanted in the main body and hopefully also helps make things more concrete for Reviewer 3. We've simply referred to the code, lodged on GitHub, to reproduce the examples in the paper, which we believe is important for transparency and reproducibility.

C7: (4) The equations in the main text are to be replaced with a narrative discussion.

Response: We agree. We have removed equations from the main text and restricted these to Box 2 and 3.

C8: (5) The Abstract & the Highlights - currently they overlap, which is a lost opportunity. Consider these as a complementary pair and work on them together. The Abstract should cover the paper at hand - focus on your key findings, not what you are going to cover. The Highlights should place the article into the broader literature. The final bullet of the Highlights is your key "take-home" message - write this first. Then the ones above move up and out to be more general and broader.

Response: We agree. We have now re-written the highlights.

C9: I have also provided a marked up copy of your manuscript with comments and suggestions that I would like you to follow. I have also uploaded the Cell Press DI form, which you must provide on resubmission. **Major (non-typographical) comments must also be included in your response to reviewers document**. To access this, please log into the Trends in Ecology and Evolution submission system at https://tree.editorialmanager.com/, click on 'Submissions Needing Revision', then the blue 'Action links' on the left. You'll see the option 'View Attachments', the relevant file should be here. If you have trouble viewing this manuscript, please contact me at tree@cell.com.

Response: Thank you. We have incorporated these suggestions into our revised manuscript as well. This includes:

- Removing highlights from the body of the text and re-writing them.
- Removed preprints.
- Trimmed outline text greatly to keep it more focused.
- Edited figure legend texts and shortened them.
- Removed the supplemental online material. We only refer to code now for reproducibility.
- Added TDT to the glossary.
- Used common names for species on first use.
- Removed equations from main text and only included them in boxes.
- Cut down words in boxes so they are ~ 400 words.
- Changed labels to roman numerals for figures in boxes (I and II).
- Cut definitions for glossary terms down.
- Cut Figure 2 legend down dramatically and moved text to main body.
- Removed Outstanding Questions from main document.
- Re-written Concluding remarks section as suggested.
- Fixed formatting issues in bibliography. Sorry about that.

Reviewer #1
C10: This paper explores how linking different types of models (i.e., biophysical, physiological, population, community, and social-ecological models) operating at various scales can inform understanding of the impacts of extreme heat on ecological systems. The authors discuss the opportunities that model integration across scales can have for understanding the impacts of extreme heat as well as the challenges involved. They touch on implications for natural and agricultural systems, food security, and ecosystem services. I think that the overall conceptual approach is appealing and offers promise.

Response: We are glad Reviewer 1 finds the conceptual approach promising and appealing for predicting how systems might be impacted by extreme heat. We thank them for their constructive comments.

C11: My concern is that there are many specific models from which to choose within each of the categories. How should one choose from among the many available models when performing such an integration? The paper would be strengthened if the authors provided some advice for how to approach this (re. quality criteria for model selection). There is a growing literature on Good Modeling Practice and guidance on model documentation, development, validation, etc. that would be helpful to briefly describe and cite. These issues become even more important when the models will be used for real-world decision-making (often by non-modelers), rather than as research tools.

Response: This is a good point. Given the breadth of models we are discussing, and the fact that we have kept them fairly high level, we cannot provide detailed advice here because what model to choose will depend on: 1) the question(s) of interest; 2) the organism(s) in question and 3) the importance one attributes to the different processes within these models. We have instead made these points clearer (**lines 287-292 & 88-90)** and referred readers to more detailed citations to help guide the model selection process, which varies across the different fields.

C12: I have mixed feelings about putting all of the case studies in the SI, which many people will not read. Including the case studies makes the proposed approach much more concrete and compelling. It may be worth considering either including a brief summary of one of the case studies in the main body of the manuscript as "proof of concept", referring to the SI for the rest. Alternatively, the case studies could potentially be published as a companion paper to the conceptual manuscript to give them more visibility and demonstrate how the proposed model integration would work in practice.

Response: This is an excellent suggestion. We agree. While we lack space to include all the examples, and already have the solutions example in the main MS, we have now included the multi-species example in **Boxes 1 & 2 in the main manuscript**. This also complements the solutions example by giving a bit more background which should help readers understand it better as well. We have also now included the TLS models more formally in this example (see **Box 2**). Including this example with detail should provide greater clarity about how such a system modelling approach can be taken when dealing with interacting species.

Some specific comments:
C13: L 54: resulting should be result.

Response: Fixed.

C14: Figure 1, L 15 of caption: "novel integration of existing models in this novel way"; probably don't need both of these novels.

Response: Thanks. We have removed the second "novel".

C15: L 208: heat injures should be heat injuries.

Response: Thanks. We've fixed this but note we've done some re-writing of this section.

C16: Box 1, first line of P 13: "can also impact thermal sensitivity" is repeated twice. Glossary, P 14, Coexistence: "A condition at odds with extreme heat events and directional climate change" is not a complete sentence. Also, delete "are" in the last line of this paragraph.

Response: Thanks for catching these errors. We've removed past text from Box 1 and fixed the glossary. We've also simplified and re-wrote the definition around "coexistence", which also removed the "are".

C17: L 232 - 233: "DEB models integrate many processes in a parameter-sparse manner" - With this I must disagree. DEB has 18 parameters, and many of them do not reflect properties that one can measure in actual organisms. This is one of the disadvantages of DEB.

Response: We have re-written this sentence, but we hold that DEB models are parameter-sparse, in that there is only one parameter per process. For example, if a DEB model is simplified to the case of a juvenile growing under constant food and temperature, it becomes a von Bertalanffy growth model with three parameters. The extra parameters come as more processes are incorporated into models via the theory (explicit embryo dynamics, fluctuating food, fluctuating temperature, ageing, maturation, reproduction …), just like any other approach to energy budgeting. The argument that DEB theory is impractical because it has many parameters that can't be estimated is frequently made but reflects a widespread and unfortunate misunderstanding of what it does and how parameters are estimated.

For further discussion see:

Kearney, M. R. 2021. What is the status of metabolic theory one century after Pütter invented the von Bertalanffy growth curve? Biological Reviews 96:557–575.

Kearney, M. R., T. Domingos, and R. Nisbet. 2015. Dynamic Energy Budget Theory: An efficient and general theory for ecology. BioScience 65:341.

C18: L 234 - 236: Although it is possible to estimate parameters without data, the jury is still out as to how robust such estimated parameters actually are.

Response: While we agree that it is best to collect data on the specific system one wishes to model, it is unlikely going to be the case that all species of interest will have complete information needed for all aspects of modelling.

We will need to rely on simplifying assumptions (as in any model building process) and data imputation approaches to build models for systems. Phylogenetically driven imputation approaches have been developed (e.g., Pottier et al. 2025; Buurgern et al. 2006) and will be useful in parameterising models where information is lacking for some species making up part of a biological system.

Additionally, there is no question that models will need to be tested empirically (and we have mentioned this in the text, lines **279-281**, **132-137 & 88-90**). While imperfect, we believe that having a quantitative model that provides clear predictions, even if these are not always correct, is better than having no model at all. Uncertainty in parameters and sensitivity of model predictions can all be incorporated in the modelling framework to capture impacts of different decisions.

We have now discussed these in more detail in the concluding remarks section (**lines 287-292)**.

References:

P. Pottier, M. R. Kearney, N. C. Wu, A. R. Gunderson, J. E. Rej, A. N. Rivera-Villanueva, P. Pollo, S. Burke, S. M. Drobniak, S. Nakagawa (2025) Vulnerability of amphibians to global warming. Nature 639, 954–961.

J. Bruggeman, J. Heringa, B. Brandt (2009) PhyloPars: estimation of missing parameter values using phylogeny. Nucleic Acids Research 37, W179–W184.

C19: L 276: Suggest to delete "ecological".

Response: Removed.

C20: Figure 4 caption: delete "and" before "compares" on line 9 and change "hazards" to "hazard" on line 21.

Response: Fixed. Thank you.

Reviewer #2

C21: This is an ambitious review offering an integrated fusion of several relatively new approaches including mechanistic microclimate modeling, biophysical modeling, dynamic energy budgeting, and population/community modeling. Either of these approaches could serve as the basis of their own review, but I appreciate this endeavor to create a blueprint for future efforts in combining these approaches to predict the consequences of extreme heat waves at the individual level and propagate these impacts to entire populations and ecosystems. I am particularly impressed by the extensive supplemental material and the example code that is offered, it is well documented and annotated. Given the wide-ranging nature of this review, however, I felt like there were sections that could probably be limited or removed and others that could use additional explanation.

Response: Thank you. We are very happy to hear the reviewer found our approach and the supplement useful and helpful. In accordance with Reviewer 1 and the Editor, we have removed the supplement, but we have brought in one of the very important examples of a three-species system to the main text which we think makes it clearer how a system modelling approach can work (see new **Boxes 1 & 2**). We have referred to the code in the paper for those interested.

C22: Although it was interesting, I felt like the section on the "current state of extreme weather modeling" (L02-L21), could be sacrificed. There is so much to unpack with GCMs that the section is underdeveloped and is likely not necessary. I feel similarly about the discussion on LSMs. I think this could be more simply a quick justification and review of mechanistic microclimate modeling. Other reviews already discuss mechanistic microclimate modeling (e.g., microclim, etc.), and I think it can be a more succinct review of the general approach and how it can be used to capture extreme temperature conditions at the organismal level.

Response: Thank you for this suggestion. We agree. We have removed the whole section on GCMs, only referring to them briefly in the microclimate section. We have also cut the text on LSMs as this was not strictly necessary here either (**lines 74-106**).

C23: By reducing the overview of GCMs and LSMs, there could be additional room for addressing other aspects of this approach that are more relevant. For example, there is very little on behavioral decision making and habitat selection, both of which are critical adaptive responses for species. Indeed, this is one of the more powerful capabilities of NicheMapper as is shown in the supplemental material (e.g., time spent foraging, nocturnal behavior, etc.), but receives scant attention in the review.

Response: We agree. We have now added more text to the biophysical ecology section to describe how behavioural decisions can be incorporated in these models (**lines 120-127**). We have also made it a clearer aspect of the example presented in **Box 1**.

C24: Similarly, there is a fair amount of space devoted to thermal load sensivitiy but was not present in the case studies. Overall, I think there could be a better balance between the approaches justified and explained in the review and the ones presented in the supplemental material.

Response: We agree. The TLS framework is very important and agree we didn't do a good enough job linking it. We have now shown how to incorporate TLS into the systems modelling approach (**Box 2**). We have also removed elements of this section to shorten it and brought it into a new box which shows the explicit process of linking models in the main text (**lines 140-169**).

C25: The biophysical modeling component, as illustrated in the supplementary material, requires a significant amount of parameterization (e.g., solar absorptivities, emmissivities, leaf orientation) as does the microclimate modeling (e.g., percent shade, etc.). I appreciate this is a general criticism (or advantage) of mechanistic modeling, but it left me wondering what types of recommendations you would offer to researchers and conservation practitioners interested in parametrizing these models with empirical data. I think this is of critical importance given your use of a socio-ecological example of agricultural systems.

Response: We have removed the socio-ecological example from the paper given the suggestion by Reviewer 3 and focused the paper mainly on the biological predictions which we think is more appropriate.

Choosing the right level of detail in any model depends on the question on hand and the processes one wants to model. Thus, it is somewhat challenging to provide clear recommendations within the context of our paper. However, we have now made this point clear and pointed to some references in various places that readers can consult to obtain a more in-depth discussion on this point (**lines 279-281**, **132-137 & 88-90**
287-292)

C26: Given the real world implications of managing these systems and their associated species during periods of extreme conditions, how can those in authority be assured that these models are realistic? What type of data would be needed at the habitat, organismal, population, and community level? What type of validation would be necessary?

Response: We have added a little more detail about the model selection process at various locations and some suggestions in the concluding remarks, emphasising the importance of model validation (see **lines 279-281**). Ultimately, we are proposing a path forward, but it's up to each research group to validate and update the models for their purpose. It's hard to give specific advice to this respect as it will depend on what the models are being used for.

Reviewer #3

C27: The consequences of extreme events including extreme heat and heat waves are truly critical for our future. These types of events are inherently challenging to forecast as well as hard to understand the consequences of because they are rare and because they involve non linear dose-response dynamics. From this perspective, the paper intents to provide inputs to a very important area. So, while I find the subject to be overly important and have deep sympathy for the intentions behind the paper, I am really sorry I cannot be more optimistic than my below review tells. I think the subject has great potential, but the paper is very difficult in its present form.

The paper is said to be an "opinion" paper which gives some freedom for more open reflections and way forward thinking for the area, and the title indicates, that the paper describes an approach to modelling impacts of extreme heat at the systems level. Maybe this is a bit challenging, because an "approach" promises or indicates a more concrete solution, and the paper does not present any such approaches.

Response: Sorry for the confusion. This paper is more of a Review as indicated in our response to the Editor. We have now tried to make the “approach” more concrete by including a hypothetical example from our supplement (see new **Boxes 1 & 2**). This example involves plants and two insect species, and we discuss how the modelling frameworks are interconnected within this context. It also should help provide a foundation for the solutions example (**Figure 2 and lines 243-271**, which we also expand).

C28: The paper describes a range of relevant and correct perspectives and examples with primary focus on heat wave related situations, but it appears more as a range of wishes, ideas and opinions rather than an "approach", and to a large extent these wishes and ideas are not put into any overall framework guiding the discussions, but rather seems to be some arbitrarily chosen examples that end up failing to present a comprehensive and holistic framework, which I would have liked and which I believe the authors have in mind. For example, the paper describes how our present understanding and modelling capacities makes it possible to model heat wave impacts across existing domains and disciplines, which is normally siloed. Likely, this is correct, but the cross boundary handling of the different aspects and challenges is nevertheless far from trivial, and the paper does not provide any operational ideas on how the different domains can interact for a comprehensive and holistic treatment. This is illustrated already by the 4 highlights at the very start of the paper which are correct, but also overly broad and generic and without any more specific suggestions for a way forward.

Response: We do not deny a systems-based approach will be challenging (**lines 273-274**). However, few papers attempt to bring together these models and show how they can be coupled and most only do this on a single species. Despite challenges it is becoming increasingly more feasible to do this with multi-species assemblages and our paper is a call to action for readers to start thinking about it more seriously given the predictive power coupling mechanistic models across species may provide.

C29: The abstract sets the stage correctly - we need to integrate models across scales. However, it further claims to show how existing models can be linked, which it does not.

Response: We have re-written aspects of the paper and brought from the supplement a clearer example (**Boxes 1 & 2**) to show more clearly how the models are connected and how to apply them across species.

C30: The introduction explains how the underlying causes and ramifications of observed impacts of climate variability and extremes are not fully understood (lines 45-53). This is clearly correct, but the paper lacks to explain how an integrated modelling approach will or can change this problem, and to provide a clear link to the ecological understanding. There is no doubt that merging and integrating modelling approaches at different scales and domains would offer a strong potential, but this does not in itself provide the understanding we miss, but rather could serve as a strong tool for generating and testing model based hypothesis and create collaboration with and identify questions and understanding to be tested through observations and experiments. Part of the problem is that our ecological understanding of extremes and their impacts is missing and requires more observations and experiments to provide better biological understanding. This interaction and collaboration between models and observations is completely missing.

Response: We have re-written large parts of the introduction (**lines 50-71**) and completely re-written our concluding remarks section to address this comment (**lines 273-295**). We have also completely re-focused our Boxes (**Boxes 1 & 2**) to demonstrate a case study which should make it clearer how such an approach provides a clear link to ecologically relevant outcomes.

C31: I do not understand the statement in lines 54-58, and the statement in line 58-59 indicates that coupling of different model concepts does not take place today. I do not think this is correct. Several of the authors are deeply involved in different model concepts that to various degrees couple different domains. It may be that it is not good enough and there is a need for more, new or better coupling, especially when we speak about extremes and heat waves, but this has to be described much more precisely.

Response: Sorry that this was confusing. We have re-worded both these sentences to hopefully make it clearer what we mean (**lines 45-52**).

C32: Line 60 - I cannot read from the context and the previous paragraph what "feat" refers to?

Response: We have re-written this section, and this comment is no longer relevant.

C33: Line 62-66 - It is unclear what the challenge with heat waves relative to general temperature increases is and how general climate change driven temperature increases and extreme heat waves are different or connected. Much of the text and descriptions sound more like impacts of general temperature increases, but heat waves have some distinct differences. What specific challenges do you see in the understanding and the modelling, at different scales, and what are you delivering or proposing? Also, how are heat waves different from other extremes? It seems to me that a modelling approach with the characteristics the authors wants to present will include some basic features, solutions or principles which have common characters for extremes, yet also for the specific case having features that are specific for temperature and heat. I would expect some degree of treating the general aspects of extremes as well as the more specific heat aspects.

Response: A system-based approach that uses physiologically explicit mechanistic models doesn't necessarily mean they only work for extremes. They capture all environmental impacts on physiology and translate them to outcomes on the life-cycle, life-history and fitness. This is the benefit of using physiological models. We have re-organised the introduction (**lines 37-71**) and re-worded these lines (**lines 49-53**) to hopefully make this clearer.

C34: Line 67-69 - it sounds like current modelling does not integrate detailed process modelling into community and population dimensions. I am sure there are important limitations that needs to be improved and developed, and that heat waves and other extremes provide much understudied cases, but I disagree that such integrations do not exist. Please do not simplify and overstate, but be precise and specific.

Response: We agree that some integrations already exist (e.g., microclimates to biophysical models), however, these do not capture thermal load sensitivity and often they are only applied at the level of a single species, not a community. Without coupling these various models for a multi-species assemblage, we cannot capture complex interactions and feedbacks between species (e.g., changes in microclimates, coexistence etc). We hope the addition of the new case study for **Box 1 & 2** helps and we have re-written the introduction (**lines 37-71**) and various other sections to emphasise these points more (e.g., **lines 74-106 & 108-137**).

C35: Line 70 - "We bring together modelling approaches ….". No, you present some loose ideas and wishes, but I do not see an approach.

Response: We have removed this statement, but we do indeed now showcase how to do this explicitly in our boxes (**Boxes 1 & 2**, plus **Figure 2**). These were in the supplement in the original version but are clearly better placed in the main text. Thank you.

C36: Line 89-92 - I think there is a confusion or mixing of general thermal adaptation and extreme heat damage. The sentence suggests that individuals mitigate extreme heat through habitat selection, but I think there is a lack of ecological description or understanding here. Most ecosystems and habitats have a history of extremes to which they are adapted. This may involve severe damage to individuals and eventual dieback from which the system may recover and over time disappeared species may return. In a future climate with novel and more severe types and intensities of extremes, these may change the pressure on individuals with dieback and avoidance, but the actual adaptation and new habitat selection is not entirely driven by the extremes, but by the general site specific bio-geo-meteorological conditions. I think the paper would be strengthened by including ecological thinking and theory much better.

Response: Throughout we have discussed the ways that extreme heat can be mitigated through behavioural plasticity (see **lines 118-130**). Also, we don't disagree that organisms will be adapted to their environment. All the points are captured by: 1) modelling how organisms behaviourally respond (which we have better shown how to do) and 2) the differences in their thermal sensitivity, which makes some species more susceptible to extreme heat compared with others (**lines 140-157**). Hopefully our revisions help address these points but if there are more specific examples the Reviewer is thinking of, we're happy to incorporate them.

C37: Line 96-98 - Is this integration actually new - or what is new in your presentation? A link to experiments and observations would be great.

Response: We have re-written this part of the introduction and tried to better emphasise what we mean by "integration" (**lines 49-53 & also Highlights**).

C38: Figure 1 - This is an extremely busy figure. It is really not easy to understand what the message or illustration is? At a first glance it seems to present general connection between basic ecological understanding and dynamics with some arbitrary or selected models, but it is unclear which boxes are models and which are system entities, it is unclear what the dotted lines indicates and it is unclear which aspects of the ecosystem functioning are involved and how they are linked. Are the ecosystem boxes separate from the modelling or are they part of the modelling (which must be the case for many of the models??). And finally, the figure claims to focus on extreme heat - but it is unclear how this is done and how it interacts with other drivers such as especially water and management. The figure is not entirely helpful and it is not clear when you refer to existing models, approaches and concepts and when you present a wish or suggestion for ssomething new.

Response: We believe this figure captures the broad classes of models we discuss and how they are interconnected so would like to keep it as an overarching introduction and guide to the review. However, we agree it could be simplified.

We now simplify by creating defined boxes, which categorise the different classes of models, reducing arrows, removing unnecessary text, and re-writing the legend to hopefully make the figure easier to follow.

C39: Line 101-121 - It is probably all true what is in here, but what is the purpose of spending 20 lines to describe weather modelling in general including the coding language? If weather and meteorological modelling is at all relevant in this context, I would suggest to focus on the aspects of meteorological modelling that are particularly challenged when it comes to extremes and specifically heat waves. I certainly agree that such meteorological modelling of extremes IS a big challenge. Long term and general climate modelling is reasonably good while the translation of the general weather patterns into local and regional extremes are really challenging. But this particular challenge is not what you describe.

Response: We agree, and this was a point brought up also by Reviewer 2. We have now removed the global climate modelling section and kept it's mention to a minimum focusing on citing already existing reviews in other sections (see **lines 74-90**).

C40: Line 122-135 - Again, I suggest you include some more specific ecological understanding here. The section looks at extremes mostly from a meteorological point of view, but this may not in itself be relevant if the ecosystems or species do not experience the pressure as extreme. Melinda Smith has a great paper where she distinguish between meteorological extremes and ecological extremes. That would be a relevant and great perspective to include.

Response: Thanks. We have completely re-written this entire section and brought a few more examples here. The section that follows ("*Translating exposure to organism temperature: biophysics to the rescue*") discusses how species may not actually experience these extremes, and how biophysical models can capture this by modelling behaviour (**lines 118-131**).

C41: Line 133-135 - I miss a reference for this latter sentence, or a more comprehensive unfolding of the non-linearity aspects. It is a surprise how little space you offer to the non-linearity aspects of extremes and how to model them, despite they are crucial characteristics of such events.

Response: Non-linearities and interactions are expected to emerge as an entire system is modelled, which is what we are advocating. We have mentioned non-linearities and interactions in the introduction and provided references for examples of how these are being incorporated in models (**lines 50-53**). We also mention them in our concluding remarks sections but have limited space to devote to it otherwise because it is an emergent property of modelling a system. The exact nature of these non-linearities will depend on the system, as Reviewer 3 understands.

C42: Line 159-161 - I do not understand what you try to say here? This sounds like what all models do (or try to do). What is the challenge you see here - is it modelling the thermal regime or the plant/system dynamics?

Response: We have re-written this section completely (**lines 91-106**). Hopefully its current configuration helps clear this up as we emphasise the challenges with modelling the feedbacks (see also **Box 3**).

C43: Line 165-217 - There is a lot about responses of organisms and almost nothing about systems. And all examples given are for individual species. You claim that the papers should go across dimensions - but you give almost no ideas to provide the systems integration. I know that systems consists of individuals, but doing species and individuals right does not automatically lead to correct treatments of systems.

Response: This is precisely the point. Most studies focus on single species and our argument in the paper is that we need to move beyond that to start modelling entire systems. We have very limited examples of truly systems-based modelling, and it has major benefits in allowing us to capture feedback between species within a system.

We have re-written our manuscript to hopefully emphasise this point better (e.g., **lines 45-53**). In addition, our Boxes now demonstrate more clearly how to go across systems (at least in a hypothetical sense, but this is a review paper after all, not an empirical study).

C44: Box 1 and Figure 2 - It is an important aspect to include - the mediation of heat waves at the systems level and through interaction with different other drivers. However, the figure is really hard to read and understand, and to me it is not enough to treat this in a box and figure, but the message has to be integrated into the text as well. You need to include the aspects of mediation into the description and not just rely on the reader being able to see the consequences of these aspects just by having it in a box or a figure. The figure text is not at all useful.

Response: Agreed. We have now completely removed this box and associated figure and rather refer to these mediators in the thermal load sensitivity section (**lines 158-169**).

C45: Glossary - hmmm - not clear why you bring this and why exactly those words are explained?

Response: TREE requires a glossary which is why we have it. We have chosen these terms because they are the key terms used within fields that may not be familiar to readers outside those fields. If there are others Reviewer 3 feels are more important that we may have missed, we are happy to add those or replace with existing terms.

C46: Line 248-253 - The interaction between temperature and water are really important, also for the case of heat waves, but I think it deserves a bit more attention in the paper.

Response: We have tried to better emphasise that water will be important by bringing in water budget models (**lines 108-112**) and discussing responses to both water and temperature (**Figure 2**). In fact, Figure 2 deals with ***both*** water and temperature within the context of identifying solutions to deal with extreme heat.

C47: Line 256 - What is the idea behind the use of "coexistence" as a key explanatory model? It is true that species interactions are relevant for describing the interaction between two or more species relative to the site conditions at a given site. But how is that relevant for the understanding of the overall consequences of extremes and heat waves? I would think coexistence is of core relevance for general site conditions and for more gradual climatic changes, while the sudden and potentially dieback provoking nature of extreme events, also described here by damage and repair, would be different from classic coexistence. Ofcourse, if future heat waves is a novel factor for a given site, the ability of a given specie and community to withstand this pressure is importance, but I do not really see how coexistence is the core logical framework to describe systems responses to heat waves (or other extremes). The authors may have a point, but it must be described.

Response: Yes, as you note, coexistence theory is more often used to understand how well species pairs will persist within sites or across gradients but that is not to say these frameworks are not also used to understand the impacts of pulse events (like heat waves) on community diversity, it just hasn't been applied in that way, to our knowledge.

Your point is a good one. Knowing if two species can technically coexist in a specific location is not that important for our framework. The coexistence framework provided by Modern Coexistence Theory (MCT), however, can and has been used to answer many more questions than that. So, it is the coexistence framework and more importantly the modelling tools to have been developed as extensions of MCT that we suggest are useful for studying community diversity when thinking about the effects of extreme heat.

As we now explain in the main text, modern modelling tools have been developed within the MCT framework to look at community diversity in whole networks based, for example of dynamical population growth models (Bimler et al. 2024), to tease apart stabilising and destabilising processes and to understanding how microclimate changes the probability that a species will succeed in a given community or not (Bowler et al. 2022). It is these extensions of MCT that we find so useful. We have now explained our rational for bringing coexistence theory into this discussion more clearly in the text (**lines 211-221**).

References:

M. D. Bimler, D. B. Stouffer, T. E. Martyn, M. M. Mayfield (2024) Plant interaction networks reveal the limits of our understanding of diversity maintenance. Ecol. Lett. 27, e14376

C. H. Bowler, C. Weiss-Lehman, I. R. Towers, M. M. Mayfield, L. G. Shoemaker (2022).
Accounting for demographic uncertainty increases predictions for species coexistence: A case study with annual plants. Ecol. Lett. 25, 1618–1628

C48: Box 2 - I think there is a time perspective missing in the description of heat waves as the driver for natural selection and evolution, and I think these two aspects are mistakenly mixed up. Extreme events may clearly drive selection of genetic variants that has the traits to survive the extreme pressure, but less so for evolution because it is a single and dramatic event. And evolution runs over longer time scales. Further, the consequence of extreme events may not be adaptation and evolution, but simply dieback, disappearence and replacement. Extreme events may cause dieback and complete removal of certain species or a complete exchange of existing species and systems with new and different species and system (e.g. desertification) meaning that survival and/or invasion may be more important. So, many of the aspects mentioned here clearly play an important role, but may only present part of the story and other relevant and equally important processes and ressponses are completely missing. For example, the equation 2 seems to be of minor importance (as also stated in the box text).

Response: Sorry, we are not quite sure what Reviewer 3 means here. Evolution can happen over ecological timescales, and extreme events have been shown to lead to rapid evolutionary change (for a recent review see: Baeckens & Donihue, 2025). We do not disagree with Reviewer 3 that that dieback is not an example of evolutionary adaptation, but this example is not one we discuss, so it is unclear how it pertains to Box 3.

Any systems modelling approach needs to eventually identify ways to incorporate eco-evolutionary feedback. Short term evolutionary shifts or plastic responses resulting from extreme climates not only depend (and impact) on selection strength but also phenotypic and genetic variance which can affect how a system responds in the future.

We agree however that this box needed to be more focused and emphasise a few of these important points better. It also needed to be shortened so we have re-written the text in this box (now **Box 3**).

References:
Baeckens, S. and Donihue, C.M., 2025. Evolutionary consequences of extreme climate events. Current Biology, 35(17), R850-R864

C49: Box 3 - I acknowledge that heat waves and other climatic extremes causing major impacts on plants and ecosystems will cascade to also have potentially significant societal impacts. However, the paper has a strange mismatch in scale with on the one hand quite some detailed description of single species and on the other hand attempts to embrace the entire global society - and these overly complex societal responses are handled by a single box and just 15 lines! This is just not convincing. The societal aspects of these changes may be really complex and involve severe impacts on anything from changes in classic provisionary ecosystems services like food, feed and fibre to economic progress, human behavior, migration and political unrest. These are really complex changes which the paper does not really acknowledge, and therefore the statement in line 292 that systems modelling can provide us with quantitative tools for predictins and decisions may in principle be correct, but we do not get any clear descriptions or approaches (which was the promise in the title of the paper) - or even a glance of what need to be scientifically and modelwise developed to get there. It may be relevant to include these aspects, but it has to be carefully described and framed, and the modelling aspects must be treated with the same carefulness as the physical-biological systems.

Response: We agree. We have now removed the SES Box from the paper and focused on the biological models. We make brief mention in the introduction (**lines 56-58**) and solutions section (**lines 243, & 265-271**) about the implications for socioecological modelling. Instead, we use this extra box bring in a concrete demonstration of how models in the major classes can be coupled across species.

C50: Figure 3 - This is an overly naïve figure that does not provide any useful information of the model approach.

Response: Agree. We have removed this figure and the SES box.

C51: Figure 4 - Even with the long descriptive text, it is a very busy figure not giving a clear idea of what it wants to tell. It seems to be focused on a very limited system with just 2 species which is a bit contradictory to the concept of the paper that speaks about integration across domains and areas. The real challenge and the promise of the paper is about the comprehensive systems view and approach. The figure does not provide any solutions to the larger systems level.

Response: We have now included more detail about the modelling approaches from the supplement in two new boxes (**Boxes 1 & 2**). Hopefully this makes the figure emerge more clearly now. We have also moved details from the legend into the main text (**lines 249-262**) and greatly simplified the **Figure 2** itself.

C52: Outstanding questions - It is great with this attempt to summarize the messages into the next step or research needs for the future. But, these questions should follow logically from the paper, which I do not think they do. I cannot identify for any of the 8 points where this is a logical conclusion from things presented and discussed in the paper. Moreover, these 8 points are quite substantial outstanding questions which speaks against the title of the paper promising the reader an approach to model consequences of heat waves. It seems to me that the authors should consider, whether they actually believes they can present a model approach, and if so, then present the approach with clear documentation and figures that present the different elements and how they are linked. Or, if such an approach is more wishful thinking and not achievable at the moment, then present an open review or "directions" paper on what we know and what we need in order to get there with clear identification of the elements we stand on and the steps of progress we should take. This should also include a better integration with ecological science to identify where we need more fundamental understanding that modelling alone cannot provide.

Response: We have now re-written and re-focused our outstanding questions to address this comment.

Highlights

- Heatwaves drive complex biological responses, yet because existing models rarely integrate the direct and indirect effects of heat across scales, our capacity to capture the dynamic feedbacks and interactions among species that drive population and community scale responses has been limited.
- A systems-thinking approach accommodates the multifaceted nature of extreme heat, enabling integration of climate, biophysical, physiological, population and community ecology models.
- By leveraging increasing capacity to estimate thermal exposure, damage, and repair at the individual level a systems-model approach can predict population and community dynamics, enabling forecasting across scales.
- We illustrate how the opportunities provided by a systems-modelling approach will rapidly advance our understanding of the impacts of extreme heat and catalyse new potential solutions to mitigate impacts.

A systems-modelling approach to predict biological responses to extreme heat

Daniel W.A. Noble[1] ‡, Margaret Mayfield[2] †, Ary Hoffmann[2] †, Zhong-Hua Chen[3,4] †, Steven J. Lade[5,6] †, Xuemei Bai[5] †, Danielle Way[7] †, Belinda Medlyn[3] †, Owen K. Atkin[7,8] †, Adrienne Nicotra[1,7] †, James M. Cook[3], Brajesh Singh[3], Alison Bentley[7], Ian Wright[3], Michael R. Kearney[2] ‡

Affiliations:

[1] Division of Ecology and Evolution, Research School of Biology, The Australian National University, Canberra ACT 2601, Australia

[2] School of BioSciences, The University of Melbourne, Australia

[3] Hawkesbury Institute for the Environment, Western Sydney University, Australia

[4] School of Agriculture, Food and Wine, Waite Research Institute, University of Adelaide, Glen Osmond, SA 5064, Australia

[5] Fenner School of Environment and Society, The Australian National University, Canberra ACT 2602, Australia

[6] Stockholm Resilience Centre, Stockholm University, Stockholm, Sweden

[7] Division of Plant Sciences, Research School of Biology, The Australian National University, Canberra ACT 2602, Australia

[8] Agrifood Innovation Institute, Australian National University, Canberra, ACT 2601, Australia

‡ Correspondence: daniel.noble@anu.edu.au & m.kearney@unimelb.edu.au

† These authors contributed equally to this work.

Keywords: plant-animal interactions, microclimate, biophysical models, thermal load sensitivity, dynamic energy budget models, coexistence theory, ecological modelling, population dynamic models

Abstract

Anthropogenic climate change is leading to more frequent and extreme heat waves. These large-scale events are radically re-shaping interactions among organisms – impacting biodiversity, community composition and ecosystem services crucial to natural systems and food security. Predicting heat wave impacts on interacting species requires an understanding of the processes driving differential exposure and sensitivity of organisms to extreme heat events in a life-cycle context. To achieve this predictive capacity, we need to integrate models across scales while capturing species-specific responses at the individual level. We review and demonstrate how existing models in disparate fields can be linked to achieve an increased understanding of how individuals and communities will respond to extreme heat, now and into the future.

A systems-modelling approach to understand the biotic impacts of heatwaves

Climate change is leading to warmer and more variable thermal environments globally [1,2]. Greater thermal variability is resulting in organisms experiencing extreme heat waves that lead to thermal stress impacting organismal growth, survival and reproduction, with cascading effects on population dynamics, species interactions, community composition and ecosystem structure and function [3,4]. Climate variability has already been linked to dramatic global declines in pollinator abundance [5] and crop yields [6–8], but the causes underlying such declines, and their ramifications through communities and society, are not fully understood. The impacts of extreme heat on species and communities are driven by a combination of direct effects of heat stress on the physiology and fitness of organisms within a given species [9–11] and indirect effects on interactions (both positive and negative) among species [e.g., 12]. Understanding the dynamic interplay between direct and indirect effects of extreme heat, and how these are mediated by environmental factors (e.g., water and food availability, microbial community), has been hampered by inadequate coupling of models that predict short-term physiological damage on a given species' fitness with population and community dynamic models across species [11,13,14]. A systems-thinking approach is now needed to tackle the multifaceted nature with which extreme heat manifests within a community. This approach will require interdisciplinary collaboration to integrate biophysical, physiological, population and community ecology processes so that we can capture and model the dynamic feedbacks, nonlinearities and interactions between species that drive responses across scales [15,16].

Using plants and insects as examples, we review broad classes of models across subfields of ecology and evolution and discuss how they can be linked to effectively model the biotic impacts of heatwaves from individuals to communities (Figure 1). We focus on plant and insect communities given the strong interconnections between them, the ease with which they can and have been studied, and their importance to socioecological systems [17]. Physiological models can now be seamlessly integrated with biophysical models to characterize temperatures experienced by organisms and simulate the effects of extreme heat events on the entire life cycle of species within ecological communities [18–21] (Figure 1). Physiological models that incorporate estimates of thermal sensitivity across species can capture the delicate balance between damage and repair of physiological systems [10,11,13,22], yielding predictions of the immediate and cumulative impacts of extreme heat on growth, survival and reproduction. Importantly, mechanistic physiological models provide outputs at the individual level (e.g., energy and water requirements, waste production, activity constraints, vital rates) that can be integrated into population and community ecology models to capture how extreme heat events perturb interactions among species and across whole, complex communities [23,24]. We demonstrate how such a coupling can be made within and across species and highlight the opportunities it presents to develop a greater understanding of the ways in which extreme heat stress manifests across species. We discuss the challenges of coupling models across diverse species in communities, and of capturing the eco-evolutionary feedbacks that will be necessary for accurate predictions in the future.

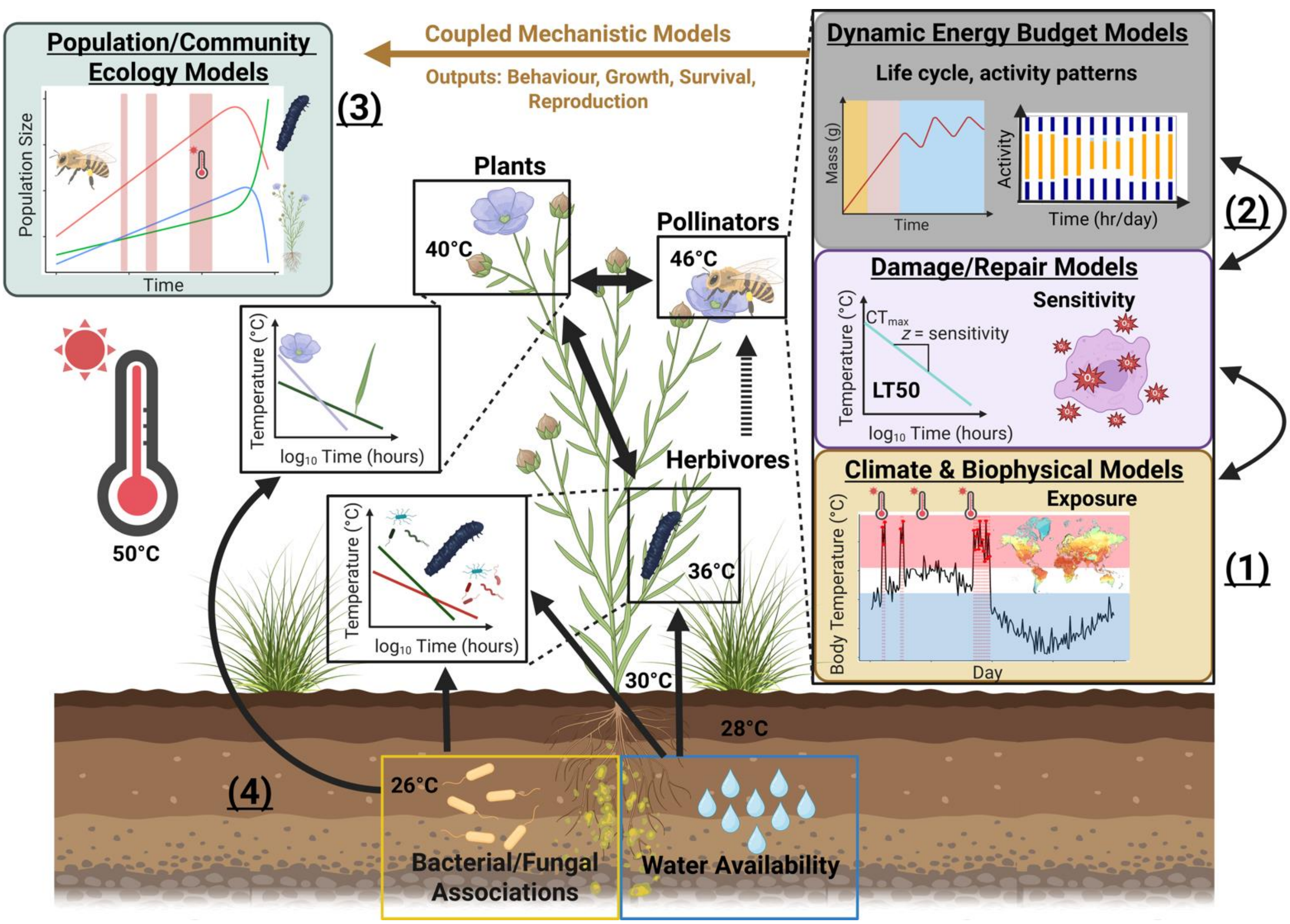


Figure 1- A coupled systems modelling framework for predicting the effects of extreme heat on ecological communities. A systems modelling approach first integrates (1) climate/biophysical models to predict the different microclimates species experience. (2) Microclimates experienced by organisms then provide inputs for physiological models that integrate temperature exposure with explicit physiological processes that simulate development, behaviour, survival (e.g., 50% survival thresholds, LT50) and reproduction in response to microclimate dynamics across life history stages. Physiological models can be built around the unique life cycles of the diverse species in communities, capturing lagged responses to extreme heat, phenological mismatches and mechanistically informed responses to extreme heat for species of the system in question. Outputs (behaviour, growth rate, biomass accumulation, survival and reproduction) from physiological models of individual organisms can then be integrated into (3) population and community ecology models (such as those developed with in the Modern Coexistence Theory Framework – see text) to predict population growth and community composition under a specific type of change (either to the environment or suite of interacting species). Environment-mediated feedback loops (4) influence organism thermal exposure and sensitivity, including water availability and microbial communities, which can vary across different life stages and organs depending on the microclimates occupied. Coupling existing models will allow for quantitative predictions to be made on how extreme heatwaves perturb biological systems and for the development and implementation of strategies to enhance biological system resilience to extreme heat. Created with BioRender.com.

From weather to microclimates: predicting community wide exposure to extreme heat

Extreme weather events such as heatwaves are predicted by Global Circulation Models (GCMs), typically at large spatial scales (e.g. 0.5°grid cells), and can be regionalized by downscaling to smaller spatial scales. However, species within communities experience heatwaves differently because of the varied microhabitats they occupy. For example, while air temperature may be 50°C, an organism a metre below the soil surface may be exposed to temperatures ~20°C cooler (Figure 1). Therefore, to understand how extreme heat affects a community of organisms, we must characterize the **microclimates** experienced by individuals of different species under a given atmospheric event. Microclimate predictions are made by taking the atmospheric conditions as independent forcing variables and combining them with detailed information on terrain (slope, aspect, hill shade), vegetation [plant-area index (PAI), stomatal behaviour, leaf reflectance] and soil hydrothermal properties to predict how radiation, wind speed, air temperature, humidity, soil temperature and soil moisture vary on small spatial scales [25–32]. Capturing the interaction between soil moisture and temperature in microclimate models is critical because the same atmospheric heat wave can have different implications for microclimatic conditions depending on the recent history of rainfall and temperature [33]. Downscaling to microclimate conditions requires the specification of key environmental properties (e.g. % shade, soil characteristics, surface albedo) at a fine scale (metres) for the local area of interest. Depending on the organism and question of interest, not all input variables are needed to parameterize a given microclimate calculation, as recent guides to microclimate modelling illustrate [e.g., 34,35,36].

Beyond parameterization, the benefit of a systems-thinking approach in the context of microclimate modelling is its ability to capture vegetation and soil dynamics which can dramatically shape microclimate conditions within and between species dynamically across time. While the large-scale feedbacks from plant and soil dynamics to heatwave development are captured in GCMs [e.g. 37], these dynamics are also important in determining thermal regimes at micro-scales. For example, plant water use dynamics can amplify or reduce within-canopy temperatures via effects on photosynthetic capacity, stomatal decoupling, cuticular conductance, leaf damage and plant mortality [20,38]. These impacts are important for the plants themselves and for thermal regimes available to other organisms. Forecasting future thermal regimes is challenging because it involves forecasting future vegetation dynamics, including changes in key vegetation properties such as leaf area index, as a function of plant growth, phenology, plant population dynamics and shifts in community composition. Despite the complexity and non-linearity of these interactions across the soil-plant-atmosphere continuum, there is existing capacity, and growing potential, to model them and determine individual-specific exposure to extreme heat. For example, a wide range of vegetation models is available for this purpose, from crop growth models that simulate growth and yield of crops over a season (e.g. Agricultural Production Systems sIMulator (APSIM); [39]) up to the dynamic global vegetation models (DGVMs) that simulate vegetation function and distribution at local to global scales [40].

Translating exposure to organism temperature: biophysics to the rescue!

Once microclimates are quantified, it is crucial to estimate the heat and water budgets of organisms to determine how microclimate variability within a system translates to realized organismal temperatures and hydration states. Heat and water budgets can be computed using a combination of species functional traits (e.g., body mass, metabolic rate, surface area, solar absorptivity) along with how heat energy and water are exchanged with the environment – dependent on the microclimate experienced. Such **biophysical models** have a long history [41,42] but have become more widely applied in ecology in the past 20 years, facilitated by developments in environmental datasets, microclimate modelling, and the emergence of high-level programming languages such as R [21]. Biophysical models of ectothermic animals make use of equations for energy and mass exchange between an organism and its environment and can account for complex radiative heat transfer and the role of evaporative water loss across surfaces [21,25,43]. Equivalent models of leaves incorporate the dynamic role of stomata [20, 44] (**Box 1**). Biophysical models are powerful because, by translating microclimate conditions to organism body temperature and hydric states, simple regulatory decision-making models can be used [45,46]. For example, given an understanding of an organism's thermal activity windows, thermal optima, and critical thermal limits we can determine what microclimates the organism should chose to get close to its target body temperature [e.g., 45,46]. Such behavioural thermoregulatory decision making is crucial for mitigating the negative impacts of extreme heat. Incorporating it into models can inherently capture the trade-offs such a strategy entails through reduced foraging time [21,45]. Similarly, changes in stomatal behaviour can mitigate or exacerbate the extreme temperatures leaf tissue experiences while simultaneously shaping the microclimate conditions for other organisms [20,47] (**Box 1**). Future developments in biophysical modelling of organismal temperature under extreme heat will involve understanding the nuances of plastic physiological and behavioral responses. For example, lizards may pant when exposed to high temperatures [48], birds may 'wind surf' or seek thermal micro-refugia [49], and stomata can enact emergency cooling, departing from the typical responses to vapor pressure and light intensity [38], though this response may depend on prior soil moisture conditions [50]. Although biophysical models require many traits for parameterization, the models and parameters can be tailored to the question and organism of interest [21]. In addition, many traits can be approximated based on traits of similar species, and then validated and updated as more species-specific data become available. The application of biophysical models in concert with microclimate models can thus be used to infer the thermal conditions to which members of the same ecological community are exposed with different degrees of detail. Realised thermal exposure can then be linked to physiological models.

Capturing both physiological damage and repair to predict multi-species thermal sensitivity

Translating how heat waves impact plants and animals not only depends on modelling temperature *exposure* but also the varying *sensitivity* of organisms to extreme heat [22,51]. Sensitivity to extreme heat can be captured by **thermal load sensitivity (TLS) / thermal death time (TDT)** models that explicitly account for how heat stress depends on both the body temperature experienced and its duration [13,22]. TLS models predict the relative accumulation of damage to cellular and sub-cellular systems that compromise physiological function. Without periods of recovery, where damage can be repaired, organisms accumulate damage over time, reducing growth and impacting survival and reproduction (**Box 2**). Typically, TLS models focus on endpoints that include survival (e.g., lethal temperatures, LT_{50} or LT_{80}) or some measure of reduced fertility, but this need not be the case [13]. Endpoints are predicted by assuming the effect of time at a given temperature decreases survival and/or fertility exponentially, and have been shown to have high predictive power [10,11,22,51]. For example, Ørsted *et al.* [11] show that both mortality and fecundity follow a clear exponential relationship with time in the Spotted Wing Drosophila (*Drosophila suzukii*), with survival and fecundity being compromised most for long thermal exposures at high temperatures. Importantly, they also show that reproductive sensitivity can be higher than mortality, with heat injury impacts accumulating faster for reproduction than survival [11]. TLS theory also applies well to photosynthetic function in plants [52,53], highlighting its generality. TLS models are crucial for predicting how heat waves affect organisms because accumulated damage to physiological systems can result in lagged responses to heat stress or exacerbate future stress (i.e., future heat waves / droughts) – a common feature of extreme heat events [13,54].

Environmental factors known to impact thermal sensitivity, such as water and drought stress [e.g., 55,56], along with nutritional and dietary changes [e.g., 57,58], can be incorporated into systems modelling approaches through their impacts on thermal sensitivity and tolerance [13] (Figure 1). Indeed, an exciting potential application of a systems-modelling approach is to explore how microclimatic conditions mediate changes to interactions between plant and animal microbial communities. Interactions between plants and microorganisms, such as plant growth-promoting rhizobacteria (PGPR), arbuscular mycorrhizal fungi (AMF), and bacterial or fungal endophytes, are known to enhance growth, defense, and heat tolerance in plants [59,60], and gut microbiota can improve heat tolerance in animals [61]. Additionally, modelling the intricate balance between damage and repair for a suite of different species can help identify susceptible species, life stages and tissues that are most at risk from extreme heat events due to direct sensitivity to extreme heat, allowing more accurate predictions of the varying levels of species sensitivity within a community [13].

Box 1: Building a systems modelling approach to capture multi-species exposure to extreme heat

Here we outline how a systems-modelling approach can be developed, illustrated with a simple hypothetical community (Figure IA). The focus is on how climate, microclimate, and biophysical models can predict the differential exposure to extreme heat events across species, how species can alter each other's microclimates, and the incorporation of 'behavioural' responses of species in response to temperature (e.g., stomatal behaviour, thermoregulation). Predicted organismal temperatures are used from these models to then predict life-cycles for species, incorporate differences in thermal sensitivity, and ultimately estimate vital rates for population and community ecology models (**Box 2**).

The context is a heatwave event in January 2018/19 at Renmark, South Australia (Calperum Station). The historical SILO climate dataset (0.05° resolution, 1889 to present) [62] was used as input to the `micro_silo` function in *NicheMapR* [20,63], in conjunction with *microclima* [26,27] in the R statistical language to compute microclimates. Microclimates are calculated at various heights above ground (20cm-1.2m, relevant to our plants and insects) and soil depths (for burrowers) with varying levels of shade (we use 40% and 80%). Simulating these varied conditions captured the diversity of microclimates that different species in a community might experience and provided opportunities to build in behavioural plasticity.

With available microclimates predicted, we can compute leaf and body temperatures using biophysical models. However, plant temperatures will impact the temperatures experienced by the grasshoppers because plant stomatal conductance changes in response to extreme heat events. Stomatal responses of leaves to their environment can be captured with the help of the *plantecophys* package [44] and used in conjunction with the `ectotherm` function in *NicheMapR* to compute realistic leaf temperatures as stomata open and close in response to temperature and vapor pressure deficit (Figure IB). The combination of microclimate, leaf traits (e.g., shape, leaf conductance, emissivity) and stomatal behaviour thereby produces realistic leaf temperature estimates during the heatwave event.

Calculated leaf temperature can then become part of the microclimate of the insects (Figure IC). The `ectotherm` function of *NicheMapR* can be used to compute insect body temperature, given the microclimate (including plant leaf temperature) and insect traits. The insect temperature calculations incorporate their capacity to thermoregulate behaviourally. For example, Species B moves to cooler parts of the plant to minimize thermal stress, when possible (Figure IA), given a range of available microclimates and using information on their thermal preferences. All code to demonstrate this process is available at: https://daniel1noble.github.io/thermal_tol_interactions/.

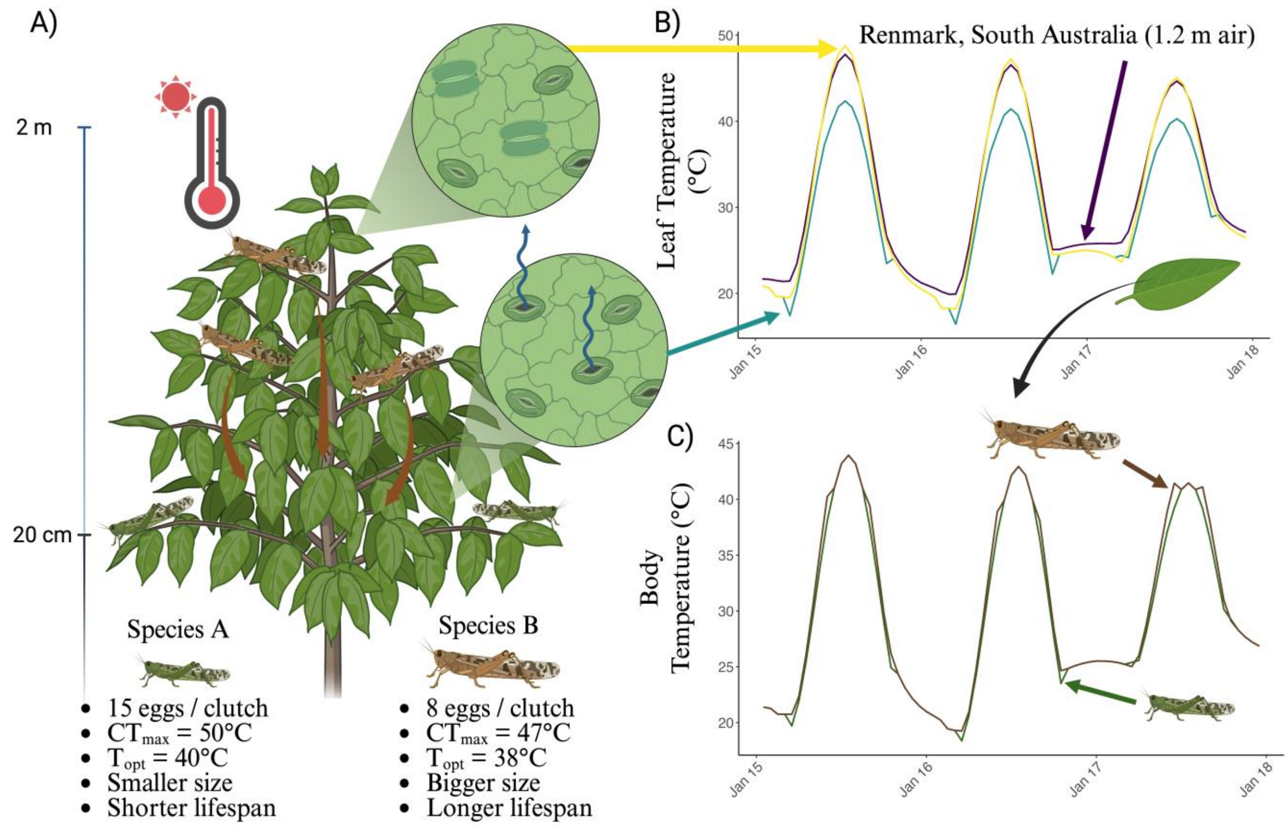


Figure I: Predicting and coupling microclimate and biophysical models of a plant and two grasshopper species that vary in their thermal tolerance and life history (A). Biophysical and thermoregulatory models use miroclimate data to predict leaf (B) and grasshopper (C) temperatures through time (coloured lines) incorporating changes in insect and stomatal behaviour. Created with BioRender.com.

169

Glossary

Biophysical models: Biophysical models capture the balances of heat, water, and other aspects of energy and mass exchange between organisms and their microclimatic environment to predict how organisms function, survive, and behave in varying environments.

Coexistence: Coexistence occurs when populations of two or more species are able to persist in each other's presence indefinitely at steady state under constant abiotic conditions.

Functional traits: Functional traits are the characteristics of organisms that influence their performance and fitness, such as body size, reproductive output, and metabolic rates. In the context of extreme heat, functional traits can be used to categorize species based on their responses to thermal stress.

Modern Coexistence Theory: A theoretical framework derived from population ecology describing the conditions under which species can coexist mathematically. It has resulted in the development of population growth models using criteria based on niche and fitness differences, mutual invasibility and feasibility domains.

Microclimate: Microclimates are the local conditions experienced by organisms, which can differ substantially from the broader atmospheric conditions. Microclimates are influenced by factors such as vegetation cover, soil properties, and topography, and can significantly affect the thermal and hydric environment experienced by organisms.

Thermal load sensitivity (TLS) / thermal death time (TDT) models: Models that calculate thermal stress load from a time series of body temperatures to quantify lethal and sublethal impacts on tissues, organs and whole organisms.

Vital rates: Demographic parameters that determine how populations change over time. They include measures of birth, death, growth, and reproduction.

Physiological models that capture the effects of extreme heat across life

Accumulated damage to an organism from experiencing extreme heat events will impact survival probability in the future and needs to be integrated with a full account of growth, development and reproduction through life to scale effects to populations and communities. Dynamic Energy Budget (DEB) models are physiologically-explicit life-cycle models of energy and mass uptake and conversion that predict key life-history transitions, growth, reproduction and survival (senescence) through time [18,64]. DEB models consider organisms as a thermodynamic system (fully obeying energy-mass conservation) capturing the exchange of food, water, respiratory gases and metabolic waste throughout the life cycle [21]. DEB models can translate short-term (hours) variation in factors such as temperature, toxins (e.g., toxicants, [65]) and resource availability (nutrients) [66,67] into long-term (days, months, years) effects on development, growth, reproduction and survival [19,21]. DEB theory, when combined with biophysical models more generally, allows the full water budget to be computed to account for heat stress impacts of a hydric nature [21]. Perhaps most importantly, DEB theory calculates the life cycle trajectory of an organism and so can account for the effects of timing of heat wave impacts relative to life stage. Applied to multi-species assemblages, DEB models can capture how different species' life cycles are affected by heat and interact with each other to predict how traits such as birth, age at maturity, reproductive events, energy flow and lifespan vary across species (**Box 2**). DEB models integrate many fundamental physiological processes and are parameter-sparse (one parameter per process). Moreover, the necessary parameters have already been estimated for > 7000 species already (although there are taxonomic biases) [68,69]. New analytical approaches using phylogenetic imputation methods show promise for calculating energy budgets for species without data. Such methods have, for instance, already been used to predict DEB parameters for over 1.3 million animal species [see, 70]. In addition, software packages such as *NicheMapR* [25,63] allow for simulations of life-cycles using parameters for diverse species. DEB model outputs, such as body mass and size, reproductive success and timing, and survival, can be used to parameterize vital rates needed for population and community models, with additional options to incorporate eco-evolutionary feedbacks (see **Box 3**).

Analogously, physiology-based vegetation models simulate plant growth over time as a function of carbon, water and nutrient uptake and use, following mass balance principles [71]. Model drivers typically include incident radiation, humidity, rainfall, and soil properties, as well as air temperature. The models capture the direct effects of temperature on key physiological processes, such as photosynthesis, respiration and phenology, as well as the indirect effects via feedbacks on vegetation water balance and soil nutrient availability. Short-term responses to temperature are captured well in these models through representations of enzyme kinetics, but representation of longer-term responses remains challenging because plants show high flexibility in their ability to acclimate to ambient temperatures [72]. Another area under active research is capturing plant damage and death from hot-dry weather extremes. Considerable progress has been made by representing the plant hydraulic system explicitly, enabling the risk of hydraulic failure to be predicted under conditions of low rainfall and high evaporative demand [73]. Direct damage to plant tissue from extreme heat has yet to be represented in such models, but the thermal death time framework outlined above offers a promising way forward.

Mechanistic approaches that 'speak' to population and community ecology under extreme heat

Modelling tools developed as part of **Modern Coexistence Theory** (MCT) [16] offer a promising suite of approaches for combining biophysical, physiological, and population ecology models to predict whole community responses to extreme heat events. Population growth models are core to predicting species **coexistence** within this framework. In their most recent applications, these simple pairwise population growth models have been used to underpin horizontal network models that allow one to examine how species in a community interact and create stable communities [15,74]. Other extensions of MCT tools have integrated environmental variation [24], traits [75] and cross-trophic dynamics [76,77] in predicting the outcomes of species interactions in community contexts. Fundamentally, this approach relies on simple two-species individual-based population growth models [78,79], but key extensions involve the use of whole communities as one "species" [80,81] or the grouping of species to simplify interaction matrices while accounting for a small number of dominant species [77].

Population dynamic models that are used as part of MCT can calculate the individual fitness effect of interactions in relation to environments experienced by organisms [23,77,82]. As these models use the same fitness measures as DEB models, they offer a population modelling framework to add thermal biology to community diversity predictions. This is because population dynamical models can be combined, for example as networks [15] and compared between microsites with different biophysical properties and harbouring species with different thermal responses to their microenvironment. For example, Bimler *et al* [15] used this approach to determine which species were keystone species under shady and sunny portions of the same plant communities.

Modified **vital rate** functions (which can incorporate any measure of fitness) can be used in population growth models to compare population dynamics for individuals in areas with different biophysical properties and/or thermal tolerances. Indeed, the predicted traits from DEB models could be used to directly parameterise growth rate models (**Box 2**) or to predict **functional traits** which, at a community scale, can be used to categorise species with different response and effect profiles. Horizontal interaction networks can then be used to determine how important microclimates with different biophysical properties are for species interactions (defined by their fitness responses to their thermal environment) within the context of whole communities. These networks can be made to target particular types of interaction effects or responses to extreme heat events, or used to compare how species interact in areas with different thermal landscapes. The main limitation of these methods is that they are data hungry, but even this can be handled by categorizing species by ‘traits’ or shared phylogenetic relationships, which are simplifications shown to be effective at reducing model complexity without sacrificing model accuracy [83].

Box 2: Scaling up extreme heat effects from impacts on individuals to populations and communities

We use predicted organismal temperatures (Figure I in Box 1) to take stock of how thermal heat stress accumulation impacts survival probability and simulate life cycles for two interacting grasshopper species under their respective microclimates.

To start, using existing parameters for a dynamic energy budget (DEB) model for grasshoppers [84], we can use the `ectotherm` function in *NicheMapR* to simulate life cycles for each of the two species. We assume that both species have similar DEB parameters but vary in their life-history, thermal physiology, size and reproduction.

DEB models incorporate a simple Gompertz mortality function to capture senescence, but do not integrate the effects of thermal load accumulation in response to extreme heat events into survival functions. We assume reproduction is not impacted by heat for simplicity, but thermal load on reproduction can also be incorporated in a similar way [11]. By using thermal sensitivity parameters for mortality endpoints from TLS models (i.e. z, slope and α = critical thermal limit), they can be used to predict the accumulation of damage through time during stressful temperatures (environmental temperatures T_e > critical temperatures T_c) experienced under realistic conditions for each species by calculating heat injury (HI) accumulated between time point t_i and t_{i+1} across the insect's life (using an LT_{50} threshold)(Equation 1) ('red lines' in Figure IIA) [10,11,13,51].

$$HI = \sum_{i=1}^{T_e > T_c} \frac{100 \cdot (t_{i+1} - t_i)}{10^{\left(-\frac{1}{z} \cdot max(T_i; T_{i+1}) + \alpha\right)}} \qquad (1)$$

After calculating HI accounting for repair [85], and converting to the probability of mortality through time, we can calculate the total probability of survival (i.e., from senescence, thermal stress, and activity-based mortality) for each age/stage class. In combination with the DEB model outputs on total reproduction across life, we can build a simple age/stage-structured population matrix (day of life or developmental stage) (Figure IIB) to estimate population growth rate [r_{max} = ln(λ)] for each species under their respective microclimates. r_{max} can then be included in a two-species, density dependent Ricker model (a model commonly used as the basis for coexistence modelling in the MCT framework) to predict their population growth under competition (Equation 2).

$$N_{t+1}^{i} = N_t^i \cdot exp\left[r_{max,i}\left(1 - \frac{N_t^i + \sum_{j \neq i} \alpha_{ij} N_t^j}{K_i}\right)\right] \qquad (2)$$

where N_t^i is the population size of species i at time t, r_{max_i} is the intrinsic growth rate of species i, K_i is the carrying capacity of species i, and α_{ij} is the competition coefficient that describes how much species j affects species i. Note that, because mechanistic physiological models are used, the population matrix can be updated each generation to reflect changes in vital rates (survival and reproduction) under different microclimates in the future.

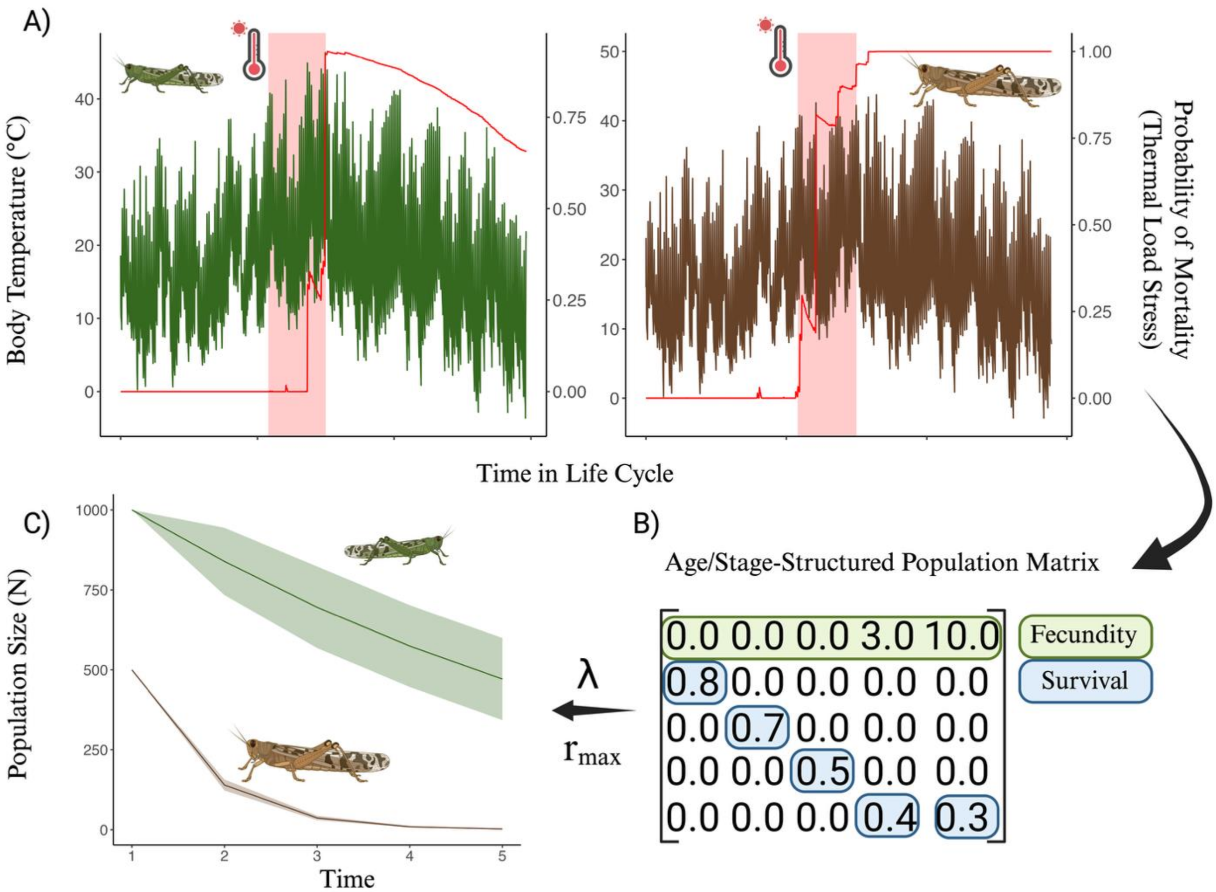


Figure II: A systems approach to using mechanistic physiological models to capture life-cycle vital rates and thermal load stress on survival to scale up to multi-species population dynamics. Created with BioRender.com.

‘Hot’ solutions for a changing world

Extreme heat will have dramatic and widespread effects on biological and socioecological systems that humans rely upon. For example, under current global climate change scenarios, each degree Celsius increase in global mean temperature is estimated to reduce the global yield of wheat by 6.0% and soybean by 3.1% [86], exacerbating food shortages.

A systems-modelling approach can help us develop suitable interventions, predict the varied consequences of such interventions, and allow for adjustments that can improve decision making for mitigating the impacts of extreme heat in real world situations. As a simple example (Figure 2A), a systems modelling approach can allow for informed predictions about how to mitigate heat stress to crops while also considering the potential for pest outbreaks. In essence, we want to add a minimal amount of water to a system to cool leaves while avoiding depleting limited water supplies and/or creating better microclimate conditions for pests. Making use of real weather data to predict microclimates and organism temperatures, we can assess how adding 2, 5, 10, or 20 mm d^{-1} of water to a system affects leaf temperatures and the life-cycle for a grasshopper species that is a known pest (Figure 2). We can see that adding 2-5 mm d^{-1} of water can decrease leaf temperature by up to 3.87°C (Figure 2B), but adding more water does not result in significant gains in cooling. In addition, 2-5 mm of water added each day results in the grasshopper population being suppressed over time, whereas not adding water at all results in grasshopper population growth, leading to potential heatwave-associated outbreaks. Here, the optimal solution is adding 2-5 mm d^{-1} of water to both cool plants and protect them from pests. Identifying and/or modifying the thermal suitability of landscapes, such as in our simple example, may dramatically benefit plants and animals under extreme heat and is a promising feature of a systems-thinking approach [21]. Additionally, as our understanding of the mechanisms associated with heat tolerance improve, genetic tools, environmental manipulations (e.g., promoting beneficial microbes) and targeted plant breeding can be used to enhance thermal tolerance, creating more resilience to extreme heat events [e.g., 60].

Systems modelling can also help understand how people's responses to the impacts of extreme heat could lead to accelerating changes in plants, animals and landscapes. Farmers or ecosystem managers may introduce new species, alter management practices or farm remaining unaffected areas more intensively. Thermal stress may ultimately result in people leaving areas, potentially making remaining communities more vulnerable through loss of resources and services [87]. Thermal risks to ecosystems may also precipitate declines in individual wellbeing, and these consequences may accrue differently across regions and peoples [88].

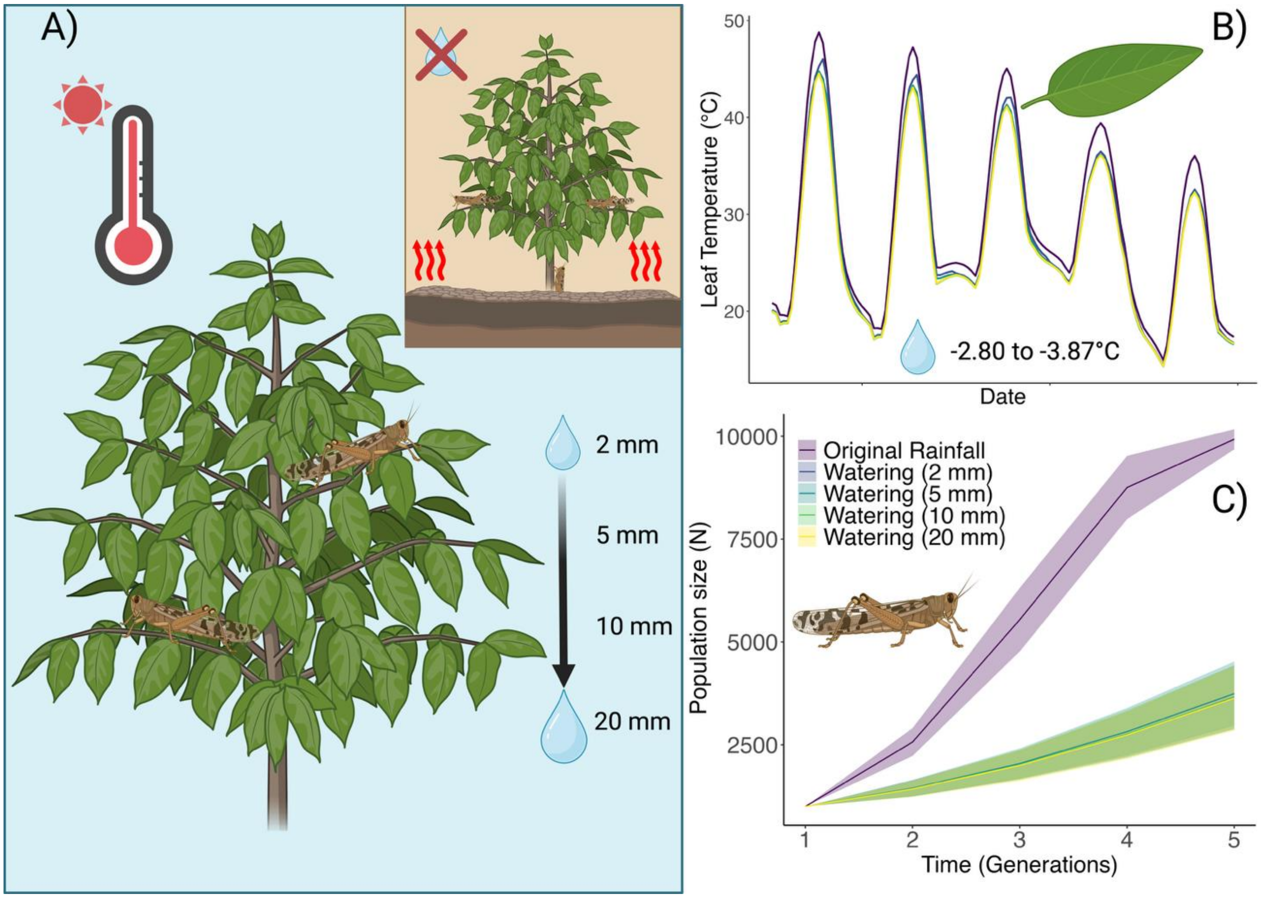


Figure 2 - How a systems modelling approach to extreme heat can be used to inform solutions. Heat stress to crops can result in foliage loss and crop failure, but also weaken crops to pests. Using a hypothetical case study of a crop and associated pest, we explore how different watering regimes affect plant temperatures and the flow on effects such regimes have for pest dynamics (A). Using a site near Renmark, South Australia, we use microclimate and biophysical models to estimate leaf temperatures (detailed in Boxes 1 & 2). We see that, compared to original rainfall conditions, certain watering regimes can significantly reduce leaf temperatures by nearly 4°C, providing protection to leaves (B). Modelling how these changed microclimate conditions translate to grasshopper body temperatures, we can then estimate changes to grasshopper life cycles and use the vital rates to predict population dynamics, showing, in this case, that even small amounts of water can indirectly suppress pest growth through lower body temperatures (C). Created with BioRender.com.

Box 3: An important frontier: integrating eco-evolutionary feedback within a systems-modelling framework

Extreme climactic events, such as heatwaves and droughts, can result in intense episodes of selection which can lead to rapid evolution of heritable traits [89]. Given the intensity of selection associated with extreme climatic events, these events have the potential to radically alter the phenotypic and genetic variability available to selection [90], potentially impacting population resilience to future events. Predicting evolutionary responses from extreme heat events is challenging because of their rarity and stochasticity which makes quantifying the strength of selection difficult as it often can only be opportunistically measured [89].

Predicting evolutionary responses also relies on our ability to predict heatwaves and how organisms experience them in the future – a task that is becoming more feasible with new climate models [e.g., 91,92]. Evolutionary consequences of extreme heat events will depend on the characteristics of these events (e.g., frequency, timing, exposure intensity and duration), the traits under selection and their levels of genetic (co)variation, as well as the demographic and ecological context in which the event occurs [93–95]. Considering various forms of plastic responses (e.g., behavioural, developmental, and physiological plasticity) as environments change through time is also crucial because plasticity can weaken selection and shelter genetic variation while also promoting persistence [96,97]. Many of these processes can now be incorporated into a systems modelling framework (e.g, behavioural plasticity), to capture environment-phenotype feedbacks.

A more complete systems modelling approach that captures eco-evolutionary dynamics needs to treat 'traits' more broadly than is typically done in evolutionary ecology (e.g., body mass). For example, 'traits' can include the parameters within models that establish functional traits within a population (e.g., DEB parameters)[98,99], which may provide predictions for suites of resulting traits that emerge from such parameters. Furthermore, it is important to consider the evolution of suites of traits as can be done using the multivariate breeders equation [100]:

$$\Delta\bar{\mathbf{z}} = \mathbf{G}\cdot\boldsymbol{\beta} \qquad (3)$$

where $\Delta\bar{\mathbf{z}}$ is the vector of changes in the mean trait values, $\mathbf{G}$ is the genetic covariance matrix and $\boldsymbol{\beta}$ is the vector of standardised selection gradients that regress each trait on relative fitness (i.e., $\boldsymbol{\omega} = \boldsymbol{\alpha} + \boldsymbol{\beta}^{\mathbf{T}}\cdot\mathbf{z} + \mathbf{e}$; [101]). Plasticity can be captured by mapping trait development to the environment and incorporating environmental variability into the breeders equation [see also, 102]. While the multivariate breeders equation can be useful for predicting short-term evolutionary responses (i.e., one or a few generations) it likely has limited predictive power over longer timespans because of changes in heritability and selection through time, which is a future challenge [95,101].

Concluding Remarks

Implementing a systems-modelling approach will no doubt be challenging, particularly for complex communities, and knowledge gaps remain (see **Outstanding Questions**). But, more than ever, we need mechanistic approaches that can capture key biological processes (physiology, behaviour, phenology, species interactions and eco-evolutionary dynamics - **Box 3**) to better predict biological responses to extreme heat [54,103]. Coupled mechanistic models within and across species are expected to have improved predictive power when projected to new environmental conditions and will better capture interacting processes [21,103]. Nonetheless, application of a systems modelling approach will require a solid foundation of species natural history, diverse modelling expertise, and a concerted effort to collect and collate trait and environmental data for model building and validation.

There are exciting new tools and data that are helping to overcome these challenges [26], and careful consideration of the key processes important to a system can help alleviate these challenges [21]. For example, large databases of physiological traits for both plants and animals now exist [e.g., 104,105] and advanced missing data approaches can help estimate likely parameters for data deficient species [70,74,106]. Ever more sophisticated and powerful computational pipelines (e.g. *NicheMapR*, *TrenchR*, *mcera5*, *terra* in R) make it easier to implement and connect models. Model selection and validation are important steps in any modelling process and will be more complicated when applied to entire systems; however, starting with a simple model, validating predictions with empirical data, and then adding complexity as needed can make building systems models more tractable [21]. Even simple models that compare counterfactual scenarios may provide important quantitative insights into the impacts of extreme heat on organisms, populations and communities [107–109]. Such insights can form the basis for prediction-driven solutions to mitigate the impacts of extreme heat on biodiversity. By working collaboratively we can develop a more quantitative and predictive understanding of the impacts that extreme heat will have on organisms, populations, and communities, and decide how to mitigate these impacts to preserve diverse systems now and into the future.

Acknowledgements

We would like to thank Pieter Arnold, Joel Treutlein and Patrice Pottier for helpful comments on previous versions of this manuscript. We would also like to thank the Editor, Andrea Stephens, and three anonymous reviewers who provided very constructive feedback on earlier versions of our manuscript. DWAN, SJL and

DW are supported by an ARC Future Fellowship (FT220100276 to DWAN, FT200100381 to SJL and FT230100193 to DW), and MRK by an ARC Laureate Fellowship (FL240100088). We would also like to thank The Australian National University, the University of Melbourne and Western Sydney University for funding to support workshops that helped develop this manuscript.

References

1. Barriopedro, D. *et al.* (2023) Heat waves: Physical understanding and scientific challenges. *Rev. Geophys.* 61, e2022RG000780

2. Perkins-Kirkpatrick, S. and Lewis, S.C. (2020) Increasing trends in regional heatwaves. *Nat. Comm.* 11, 3357

3. Evans, M.E.K. *et al.* (2025) Scaling plant responses to heat: From molecules to the biosphere. *Science* 388, 1167–1173

4. Margalef-Marrase, J. *et al.* (2020) Relationship between heatwave-induced forest die-off and climatic suitability in multiple tree species. *Glob. Chang. Biol.* 26, 3134–3146

5. Kazenel, M.R. *et al.* (2024) Heat and desiccation tolerances predict bee abundance under climate change. *Nature* 628, 342–348

6. Zhao, C. *et al.* (2017) Temperature increase reduces global yields of major crops in four independent estimates. *Proc. Natl. Acad. Sci. U. S. A.* 114, 9326–9331

7. Tito, R. *et al.* (2018) Global climate change increases risk of crop yield losses and food insecurity in the tropical andes. *Glob. Chang. Biol.* 24, e592–e602

8. Bernacchi, C.J. *et al.* (2025) Safeguarding crop photosynthesis in a rapidly warming world. *Science* 388, 1153–1160

9. Martinez, L.D. *et al.* (2023) Effect of simulated heat waves on the behaviour of two mirid predators. *J. Appl. Entomol.* 147, 486–498

10. Rezende, E.L. *et al.* (2020) Predicting temperature mortality and selection in natural *Drosophila* populations. *Science* 369, 1242–1245

11. Ørsted, M. *et al.* (2024) Thermal limits of survival and reproduction depend on stress duration: A case study of *Drosophila suzukii*. *Ecol. Lett.* 27, e14421

12. López-Goldar, X. *et al.* (2024) Heat waves induce milkweed resistance to a specialist herbivore via increased toxicity and reduced nutrient content. *Plant Cell Environ.* 47, 4530–4542

13. Arnold, P.A. *et al.* (2025) A framework for modelling thermal load sensitivity across life. *Glob. Chang. Biol.* 31, e70315

14. Trivedi, P. *et al.* (2022) Plant–microbiome interactions under a changing world: Responses, consequences and perspectives. *New Phytol.* 234, 1951–1959

15. Bimler, M.D. *et al.* (2024) Plant interaction networks reveal the limits of our understanding of diversity maintenance. *Ecol. Lett.* 27, e14376

16. Chesson, P. (2000) General theory of competitive coexistence in spatially-varying environments. *Theor. Popul. Biol.* 58, 211–237

17. Dicks, L.V. *et al.* (2021) A global-scale expert assessment of drivers and risks associated with pollinator decline. *Nat. Ecol. Evol.* 5, 1453–1461

18. Kooijman, S.A.L.M. (2010) Dynamic energy budget theory for metabolic organisation, Cambridge University Press.

19. Kearney, M. and Porter, W. (2009) Mechanistic niche modelling: Combining physiological and spatial data to predict species' ranges. *Ecol. Lett.* 12, 334–350

20. Kearney, M.R. and Leigh, A. (2024) Fast, accurate and accessible calculations of leaf temperature and its physiological consequences with NicheMapR. *Methods Ecol. Evol.* 15, 1516–1531

21. Briscoe, N.J. *et al.* (2023) Mechanistic forecasts of species responses to climate change: The promise of biophysical ecology. *Glob. Chang. Biol.* 29, 1451–1470

22. Rezende, E.L. *et al.* (2014) Tolerance landscapes in thermal ecology. *Funct. Ecol.* 28, 799–809

23. Bimler, M.D. *et al.* (2018) Accurate predictions of coexistence in natural systems require the inclusion of facilitative interactions and environmental dependency. *J. Ecol.* 106, 1839–1852

24. Bowler, C.H. *et al.* (2022) Accounting for demographic uncertainty increases predictions for species coexistence: A case study with annual plants. *Ecol. Lett.* 25, 1618–1628

25. Kearney, M.R. and Porter, W.P. (2017) NicheMapR – an R package for biophysical modelling: The microclimate model. *Ecography (Cop.)* 40, 664–674

26. Maclean, I.M.D. *et al.* (2019) Microclima: An R package for modelling meso- and microclimate. *Methods Ecol. Evol.* 10, 280–290

27. Kearney, M.R. *et al.* (2020) A method for computing hourly, historical, terrain-corrected microclimate anywhere on earth. *Methods Ecol. Evol.* 11, 38–43

28. Maclean, I.M.D. *et al.* (2021) On the measurement of microclimate. *Methods Ecol. Evol.* 12, 1397–1410

29. Buckley, L.B. *et al.* (2023) TrenchR: An R package for modular and accessible microclimate and biophysical ecology. *PLOS Clim* 2, e0000139

30. Bramer, I. *et al.* (2018) Advances in monitoring and modelling climate at ecologically relevant scales. *Adv. Ecol. Res.* 58, 101–161

31. Levy, O. *et al.* (2016) A dynamically downscaled projection of past and future microclimates. *Ecology* 97, 1888–1888

32. Choi, F. *et al.* (2019) Mapping physiology: Biophysical mechanisms define scales of climate change impacts. *Conserv. Physiol.* 7, coz028

33. Lorenz, R. *et al.* (2010) Persistence of heat waves and its link to soil moisture memory. *Geophys. Res. Lett.* 37, L09703

34. De Frenne, P. *et al.* (2024) Ten practical guidelines for microclimate research in terrestrial ecosystems. *Methods Ecol. Evol.* 16, 269-294

35. Meyer, A.V. *et al.* (2023) A guide and tools for selecting and accessing microclimate data for mechanistic niche modeling. *Ecosphere* 14, e4506

36. Klinges, D.H. *et al.* (2025) Matching climate to biological scales. *Trends Ecol. Evol.* in press

37. Hirsch, A.L. *et al.* (2019) Amplification of australian heatwaves via local land-atmosphere coupling. *J. Geophys. Res.* 124, 13625–13647

38. Drake, J.E. *et al.* (2018) Trees tolerate an extreme heatwave via sustained transpirational cooling and increased leaf thermal tolerance. *Glob. Chang. Biol.* 24, 2390–2402

39. Holzworth, D. *et al.* (2018) APSIM next generation: Overcoming challenges in modernising a farming systems model. *Environ. Model. Softw.* 103, 43–51

40. Medlyn, B.E. *et al.* (2011) Reconciling the optimal and empirical approaches to modelling stomatal conductance. *Glob. Chang. Biol.* 17, 2134–2144

41. Porter, W.P. *et al.* (1973) Behavioral implications of mechanistic ecology : Thermal and behavioral modeling of desert ectotherms and their microenvironment. *Oecologia* 13, 1–54

42. Porter, W.P. and Gates, D.M. (1969) Thermodynamic equilibria of animals with environment. *Ecol. Monogr.* 39, 227–244

43. Riddell, E.A. *et al.* (2017) Physical calculations of resistance to water loss improve predictions of species range models. *Ecol. Monogr.* 87, 21–33

44. Duursma, R.A. (2015) Plantecophys–an R package for analysing and modelling leaf gas exchange data. *PLoS One* 10, e0143346

45. Wild, K.H. *et al.* (2025) Climate change and the cost-of-living squeeze in desert lizards. *Science* 387, 303–309

46. Briscoe, N.J. *et al.* (2022) Too hot to hunt: Mechanistic predictions of thermal refuge from cat predation risk. *Conserv. Lett.* 15, e12906

47. Duursma, R.A. *et al.* (2019) On the minimum leaf conductance: Its role in models of plant water use, and ecological and environmental controls. *New Phytol.* 221, 693–705

48. Loughran, C.L. and Wolf, B.O. (2020) The functional significance of panting as a mechanism of thermoregulation and its relationship to the critical thermal maxima in lizards. *J. Exp. Biol.* 223, jeb224139

49. Sharpe, L.L. *et al.* (2022) In the hot seat: Behavioral change and old-growth trees underpin an australian songbird's response to extreme heat. *Front. Ecol. Evol.* 10, 813567

50. Posch, B.C. *et al.* (2024) Intensive leaf cooling promotes tree survival during a record heatwave. *Proc. Natl. Acad. Sci. U. S. A.* 121, e2408583121

51. Ørsted, M. *et al.* (2022) Finding the right thermal limit: A framework to reconcile ecological, physiological and methodological aspects of CTmax in ectotherms. *J. Exp. Biol.* 225, jeb244514

52. Jagadish, S.V.K. *et al.* (2021) Plant heat stress: Concepts directing future research. *Plant Cell Environ.* 44, 1992–2005

53. Faber, A.H. *et al.* (2024) Application of the thermal death time model in predicting thermal damage accumulation in plants. *J. Exp. Bot.* 75, 3467–3482

54. Martínez-De León, G. and Thakur, M.P. (2024) Ecological debts induced by heat extremes. *Trends Ecol. Evol.* 39, 1024–1034

55. Marchin, R.M. *et al.* (2022) Extreme heat increases stomatal conductance and drought-induced mortality risk in vulnerable plant species. *Glob. Chang. Biol.* 28, 1133–1146

56. Youngblood, J.P. *et al.* (2025) Dehydration worsens heat tolerance of locusts and amplifies predicted impacts of climate change. *Funct. Ecol.* 39, 1194-1207

57. Mengutay, M. *et al.* (2013) Adequate magnesium nutrition mitigates adverse effects of heat stress on maize and wheat. *Plant Soil* 368, 57–72

58. Andersen, L.H. *et al.* (2010) Protein and carbohydrate composition of larval food affects tolerance to thermal stress and desiccation in adult drosophila melanogaster. *J. Insect Physiol.* 56, 336–340

59. Chouhan, D. *et al.* (2023) Plant-microbe interaction and their role in mitigation of heat stress. In *Rhizosphere biology*, pp. 127–147, Springer Nature Singapore

60. Márquez, L.M. *et al.* (2007) A virus in a fungus in a plant: Three-way symbiosis required for thermal tolerance. *Science* 315, 513–515

61. Hoang, K.L. *et al.* (2019) The effects of bacillus subtilis on caenorhabditis elegans fitness after heat stress. *Ecol. Evol.* 9, 3491–3499

62. Jeffrey, S.J. *et al.* (2001) Using spatial interpolation to construct a comprehensive archive of australian climate data. *Environ. Model. Softw.* 16, 309–330

63. Kearney, M.R. and Porter, W.P. (2020) NicheMapR – an R package for biophysical modelling: The ectotherm and dynamic energy budget models. *Ecography (Cop.)* 43, 85–96

64. Nisbet, R. *et al.* (2000) From molecules to ecosystems through dynamic energy budget models. *Journal of Animal Ecology* 69, 913–926

65. Jager, T. (2020) Revisiting simplified DEBtox models for analysing ecotoxicity data. *Ecol. Modell.* 416, 108904

66. Kearney, M.R. *et al.* (2013) Balancing heat, water and nutrients under environmental change: A thermodynamic niche framework. *Funct. Ecol.* 27, 950–966

67. Kearney, M. *et al.* (2010) Modelling the ecological niche from functional traits. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 365, 3469–3483

68. Marques, G.M. *et al.* (2018) The AmP project: Comparing species on the basis of dynamic energy budget parameters. *PLoS Comput. Biol.* 14, e1006100

69. Kooijman, S.A.L.M. *et al.* (2021) Multidimensional scaling for animal traits in the context of dynamic energy budget theory. *Conserv. Physiol.* 9, coab086

70. Bruggeman, J. *et al.* (2009) PhyloPars: Estimation of missing parameter values using phylogeny. *Nucleic Acids Research* 37, W179–W184

71. Bonan, G. (2019) Climate change and terrestrial ecosystem modeling. Cambridge University Press

72. Kumarathunge, D.P. *et al.* (2019) Acclimation and adaptation components of the temperature dependence of plant photosynthesis at the global scale. *New Phytol.* 222, 768–784

73. De Kauwe, M.G. *et al.* (2020) Identifying areas at risk of drought-induced tree mortality across south-eastern australia. *Glob. Chang. Biol.* 26, 5716–5733

74. Bimler, M.D. *et al.* (2023) Estimating interaction strengths for diverse horizontal systems using performance data. *Methods Ecol. Evol.* 14, 968–980

75. Kraft, N.J.B. *et al.* (2015) Plant functional traits and the multidimensional nature of species coexistence. *Proc. Natl. Acad. Sci. U. S. A.* 112, 797–802

76. Buche, L. *et al.* (2024) Multitrophic higher-order interactions modulate species persistence. *Am. Nat.* 203, 458–472

77. Buche, L. *et al.* (2025) A continuum from positive to negative interactions drives plant species' performance in a diverse community. *Ecol. Lett.* 28, e70059

78. Ricker, W.E. (1954) Stock and recruitment. *J. Fish. Res. Board Can.* 11, 559–623

79. Beverton, R.J.H. and Holt, S. (2012) On the dynamics of exploited fish populations. Springer

80. Levine, J.M. and HilleRisLambers, J. (2009) The importance of niches for the maintenance of species diversity. *Nature* 461, 254–257

81. Wainwright, C.E. *et al.* (2019) Distinct responses of niche and fitness differences to water availability underlie variable coexistence outcomes in semi-arid annual plant communities. *J. Ecol.* 107, 293–306

82. Mayfield, M. and Stouffer, D.B. (2017) Higher-order interactions capture unexplained complexity in diverse communities. *Nat. Ecol. Evol.* 1, 0062

83. Martyn, T.E. *et al.* (2021) Identifying “useful” fitness models: Balancing the benefits of added complexity with realistic data requirements in models of individual plant fitness. *Am. Nat.* 197, 415–433

84. Llandres, A.L. *et al.* (2015) A dynamic energy budget for the whole life-cycle of holometabolous insects. *Ecol. Monogr.* 85, 353–371

85. Arnold, P.A. *et al.* (2019) Sparse evidence for selection on phenotypic plasticity in response to temperature. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 374, 20180185

86. Shekhawat, K. *et al.* (2022) Beat the heat: Plant- and microbe-mediated strategies for crop thermotolerance. *Trends Plant Sci.* 27, 802–813

87. Dandy, J. *et al.* (2019) Leaving home: Place attachment and decisions to move in the face of environmental change. *Reg. Environ. Change* 19, 615–620

88. Leviston, Z. *et al.* (2018) Linkages between ecosystem services and human wellbeing: A nexus webs approach. *Ecol. Indic.* 93, 658–668

89. Baeckens, S. and Donihue, C.M. (2025) Evolutionary consequences of extreme climate events. *Curr. Biol.* 35, R850–R864

90. Urban, M.C. (2024) Climate change extinctions. *Science* 386, 1123–1128

91. Muñoz-Sabater, J. *et al.* (2021) ERA5-land: A state-of-the-art global reanalysis dataset for land applications. *Earth Syst. Sci. Data* 13, 4349–4383

92. Eyring, V. *et al.* (2016) Overview of the coupled model intercomparison project phase 6 (CMIP6) experimental design and organization. *Geosci. Model Dev.* 9, 1937–1958

93. Chevin, L.-M. and Hoffmann, A.A. (2017) Evolution of phenotypic plasticity in extreme environments. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 372, 20160138

94. Hoffmann, A.A. and Bridle, J. (2022) The dangers of irreversibility in an age of increased uncertainty: Revisiting plasticity in invertebrates. *Oikos* 2022, e08715

95. Hoffmann, A.A. and Sgrò, C.M. (2011) Climate change and evolutionary adaptation. 470, 479–485

96. Catullo, R.A. *et al.* (2019) The potential for rapid evolution under anthropogenic climate change. *Curr. Biol.* 29, R996–R1007

97. Ghalambor, C. *et al.* (2015) Non-adaptive plasticity potentiates rapid adaptive evolution of gene expression in nature. *Nature* 525, 372–375

98. Sousa, T. *et al.* (2010) Dynamic energy budget theory restores coherence in biology. *Philos. Trans. R. Soc. Lond. B Biol. Sci.* 365, 3413–3428

99. Kearney, M. *et al.* (2009) Integrating biophysical models and evolutionary theory to predict climatic impacts on species' ranges: The dengue mosquito aedes aegypti in australia. *Funct. Ecol.* 23, 528–538

100. Lande, R. (1979) Quantitative genetic analysis of multivariate evolution, applied to brain:body size allometry. *Evolution* 33, 402–416

101. Walsh, B. and Lynch, M. (2018) Evolution and selection of quantitative traits. Oxford University Press

102. McGlothlin, J. and Galloway, L. (2014) The contribution of maternal effects to selection response: An empirical test of competing models. *Evol.* 68, 549–558

103. Urban, M.C. *et al.* (2016) Improving the forecast for biodiversity under climate change. *Science* 353, aad8466–aad8466

104. Kattge, J. *et al.* (2020) TRY plant trait database - enhanced coverage and open access. *Glob. Chang. Biol.* 26, 119–188

105. Leiva, F.P. *et al.* (2025) ShareTrait: Towards interoperable and reusable individual trait-based data in ectotherms. *Funct. Ecol.* 39, 3124-3138

106. Weiss-Lehman, C.P. *et al.* (2022) Disentangling key species interactions in diverse and heterogeneous communities: A bayesian sparse modelling approach. *Ecol. Lett.* 25, 1263–1276

107. Dudney, J. *et al.* (2025) A causal inference framework for climate change attribution in ecology. *Ecol. Lett.* 28, e70192

108. Carroll, T. *et al.* (2023) Biodiversity change under adaptive community dynamics. *Glob. Chang. Biol.* 29, 3525–3538

109. Dornhaus, A. *et al.* (2022) How can we fully realize the potential of mathematical and biological models to reintegrate biology? *Integr. Comp. Biol.* 61, 2244–2254

Outstanding Questions

1. How effective is using parameters of closely related species for predicting the impacts of extreme heat for data deficient species?
2. How can we best capture rapid and longer-term evolutionary responses of interacting species to extreme heat in a systems-modelling framework?
3. How can we capture multiple anthropogenic stressors in a systems-modelling framework to explore how they interact with extreme heat to affect organisms, populations and communities?
4. How will extreme heat affect species coexistence and functional trait diversity in natural and agricultural systems?
5. How do we best model lagged effects of extreme heat events on whole communities?
6. How can we extend a systems modelling approach to understand how thermal stress to natural ecosystems and food production systems impact people?

Figure 1 Click here to access/download;Figure;Fig1_rev1.png

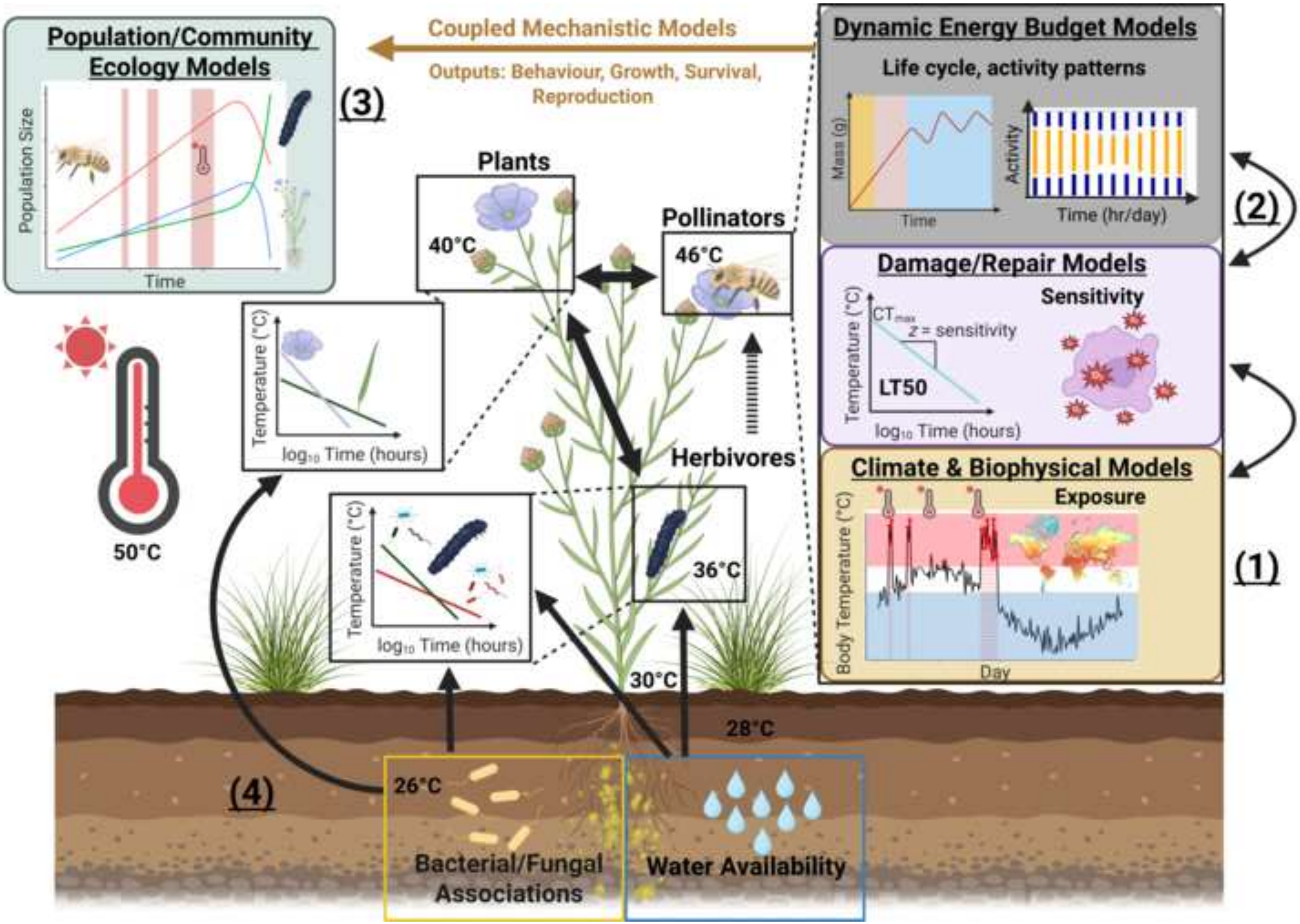


Figure 2

Click here to access/download;Figure;Fig 2_rev1.png

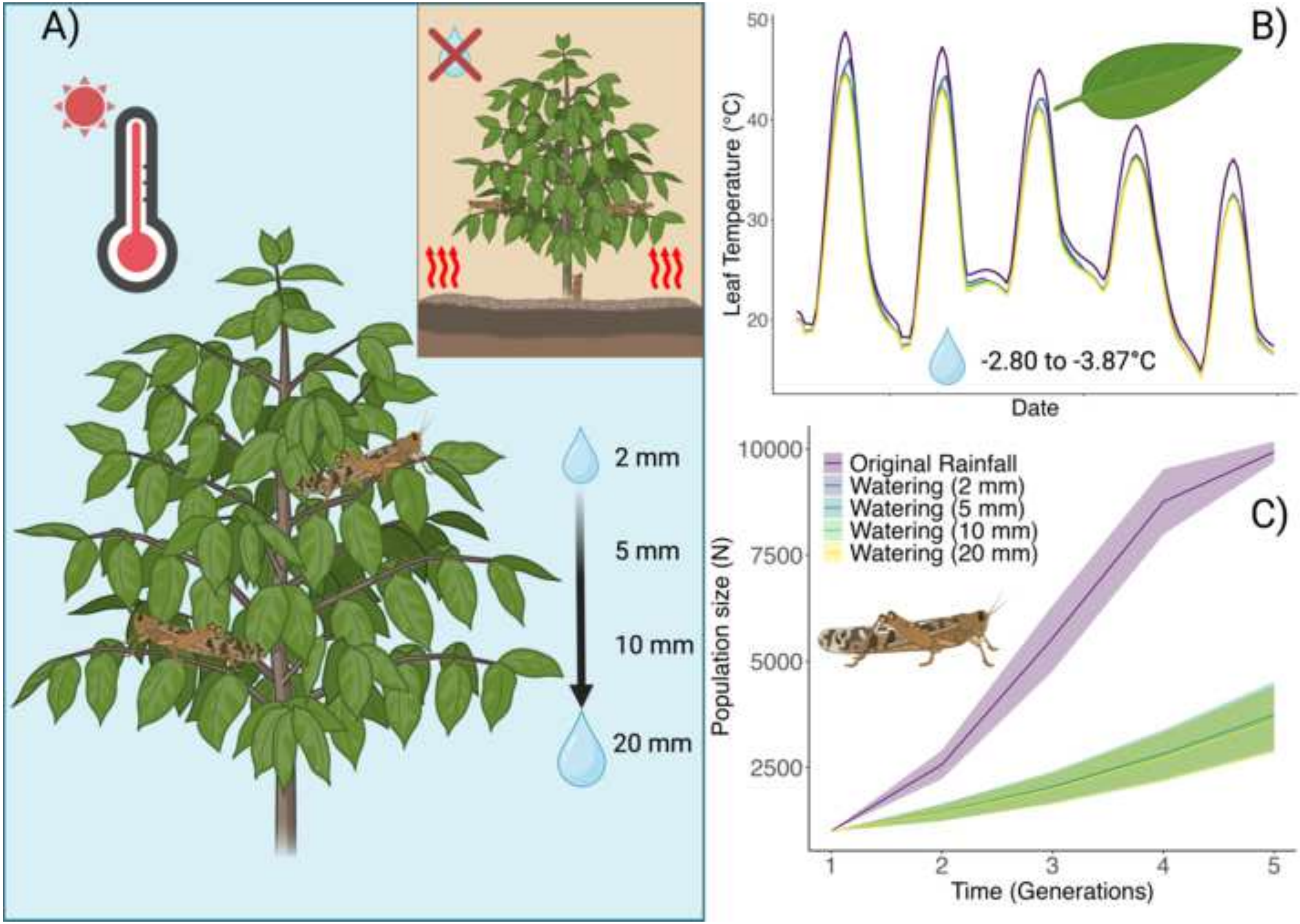


Figure I

Click here to access/download;Figure;Box1_fig.png

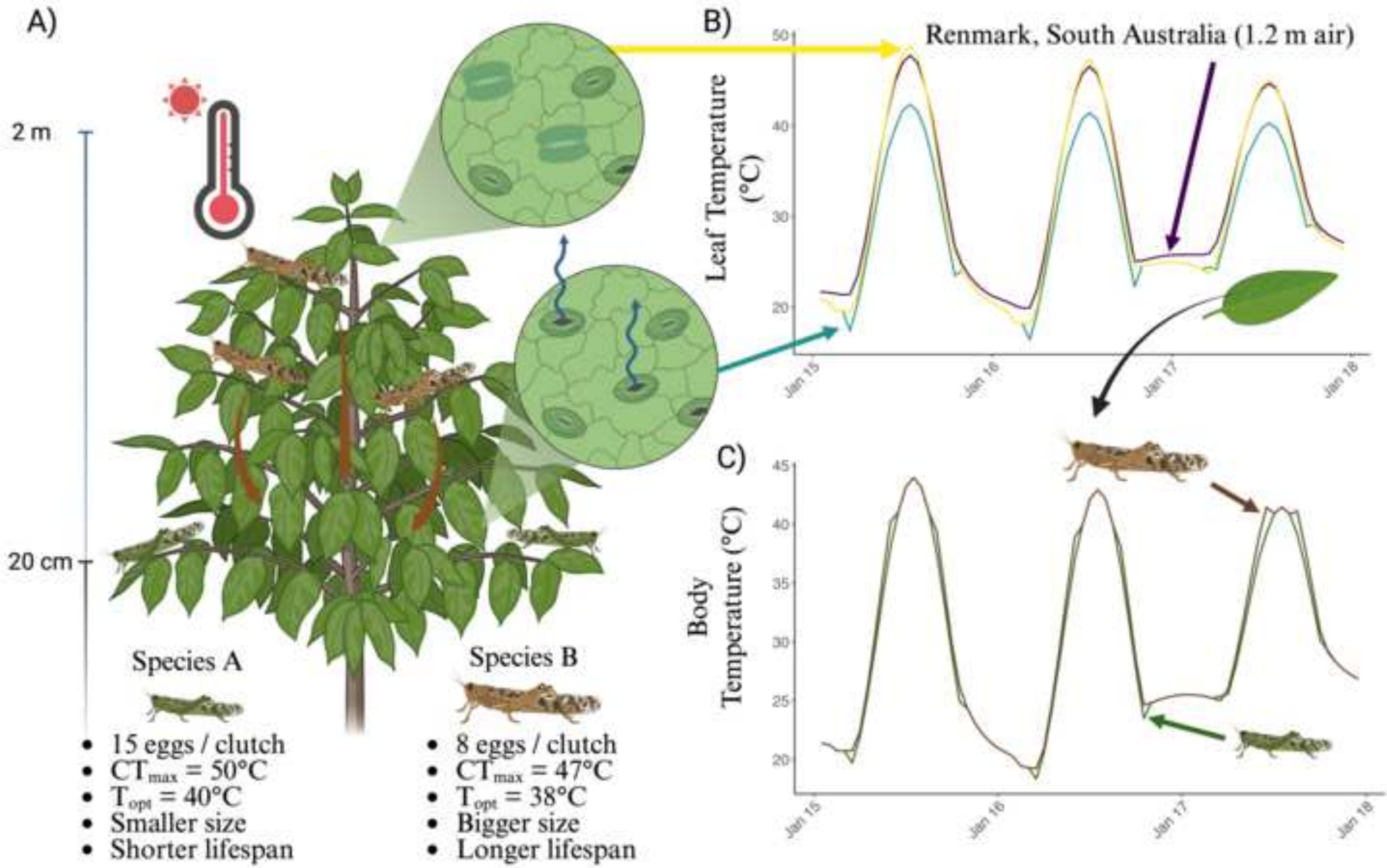
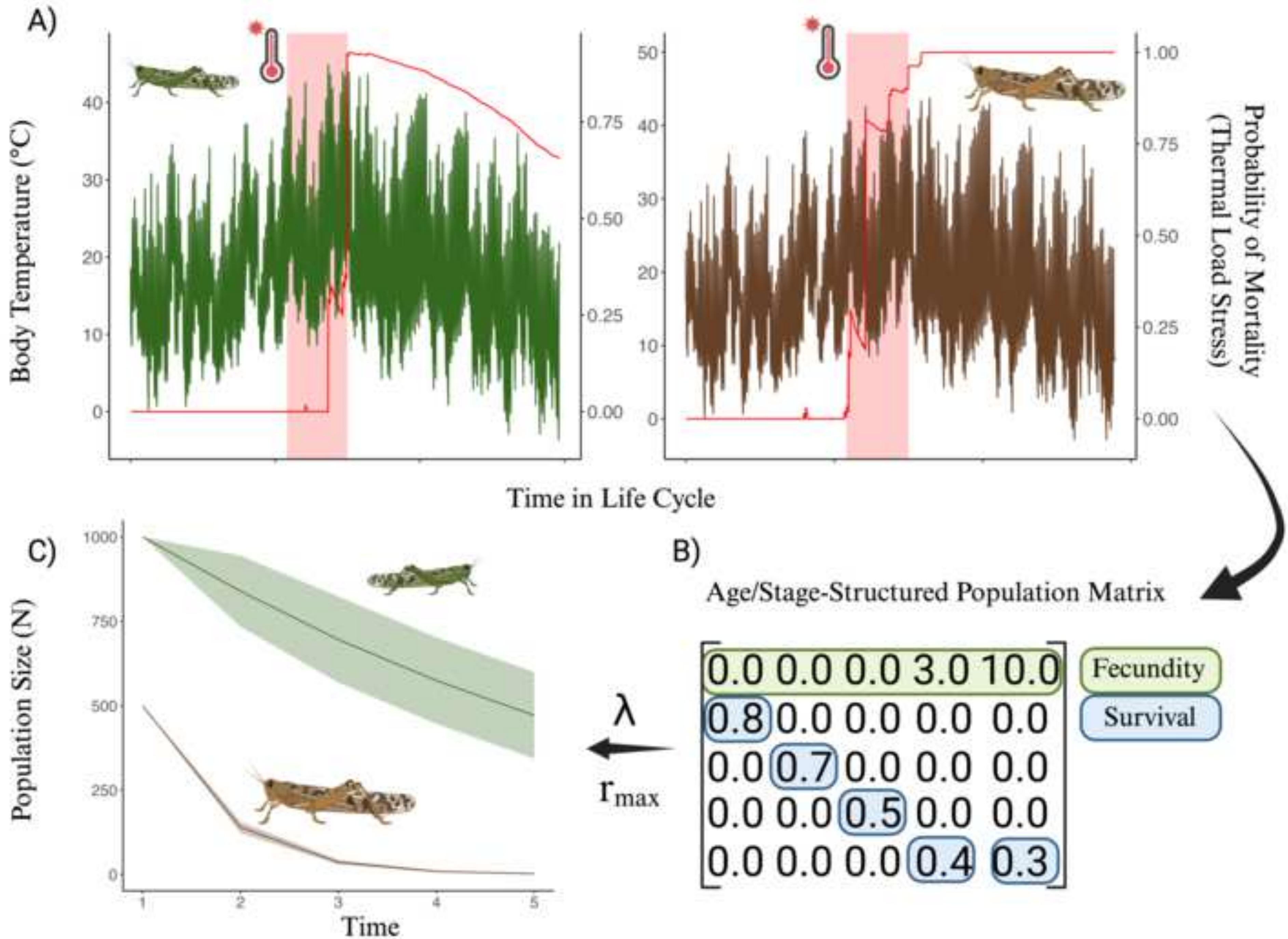


Figure II

Click here to access/download;Figure;Fig_Box2.png



CELL PRESS DECLARATION OF INTERESTS POLICY

Transparency is essential for a reader's trust in the scientific process and for the credibility of published articles. At Cell Press, we feel that disclosure of competing interests is a critical aspect of transparency. Therefore, we require a "declaration of interests" section in which all authors disclose any financial or other interests related to the submitted work that (1) could affect or have the perception of affecting the author's objectivity or (2) could influence or have the perception of influencing the content of the article.

<u>What types of articles does this apply to?</u>

We require that you disclose competing interests for all submitted content by completing and submitting the form below. We also require that you include a "declaration of interests" section in the text of all articles even if there are no interests to declare.

<u>What should I disclose?</u>

We require that you and all authors disclose any personal financial interests (e.g., stocks or shares in companies with interests related to the submitted work or consulting fees from companies that could have interests related to the work), professional affiliations, advisory positions, board memberships (including membership on a journal's advisory board when publishing in that journal), or patent holdings that are related to the subject matter of the contribution. As a guideline, you need to declare an interest for (1) any affiliation associated with a payment or financial benefit exceeding $10,000 p.a. or 5% ownership of a company or (2) research funding by a company with related interests. You do not need to disclose diversified mutual funds, 401ks, or investment trusts.

Authors should also disclose relevant financial interests of immediate family members. Cell Press uses the Public Health Service definition of "immediate family member," which includes spouse and dependent children.

<u>Where do I declare competing interests?</u>

Competing interests should be disclosed on this form as well as in a "declaration of interests" section in the manuscript. This section should include financial or other competing interests as well as affiliations that are not included in the author list. Examples of "declaration of interests" language include:

> "AUTHOR is an employee and shareholder of COMPANY."
> "AUTHOR is a founder of COMPANY and a member of its scientific advisory board."

NOTE: Primary affiliations should be included with the author list and do not need to be included in the "declaration of interests" section. Funding sources should be included in the "acknowledgments" section and also do not need to be included in the "declaration of interests" section. (A small number of front-matter article types do not include an "acknowledgments" section. For these articles, reporting of funding sources is not required.)

<u>What if there are no competing interests to declare?</u>

If you have no competing interests to declare, please note that in the "declaration of interests" section with the following wording:

> "The authors declare no competing interests."

August 1, 2023

CELL PRESS DECLARATION OF INTERESTS FORM

If submitting materials via Editorial Manager, please complete this form and upload with your initial submission. Otherwise, please email as an attachment to the editor handling your manuscript.

Please complete each section of the form and insert any necessary "declaration of interests" statement in the text box at the end of the form. A matching statement should be included in a "declaration of interests" section in the manuscript.

<u>Institutional affiliations</u>

We require that you list the current institutional affiliations of all authors, including academic, corporate, and industrial, on the title page of the manuscript. ***Please select one of the following:***

☒ All affiliations are listed on the title page of the manuscript.

☐ I or other authors have additional affiliations that we have noted in the "declaration of interests" section of the manuscript and on this form below.

<u>Funding sources</u>

We require that you disclose all funding sources for the research described in this work. ***Please confirm the following:***

☒ All funding sources for this study are listed in the "acknowledgments" section of the manuscript.*

*A small number of front-matter article types do not include an "acknowledgments" section. For these, reporting funding sources is not required.

<u>Competing financial interests</u>

We require that authors disclose any financial interests and any such interests of immediate family members, including financial holdings, professional affiliations, advisory positions, board memberships, receipt of consulting fees, etc., that:

(1) could affect or have the perception of affecting the author's objectivity, *or*
(2) could influence or have the perception of influencing the content of the article.

Please select one of the following:

☒ We, the authors and our immediate family members, have no financial interests to declare.

☐ We, the authors, have noted any financial interests in the "declaration of interests" section of the manuscript and on this form below, and we have noted interests of our immediate family members.

Advisory/management and consulting positions

We require that authors disclose any position, be it a member of a board or advisory committee or a paid consultant, that they have been involved with that is related to this study. We also require that members of our journal advisory boards disclose their position when publishing in that journal. ***Please select one of the following:***

☒ We, the authors and our immediate family members, have no positions to declare and are not members of the journal's advisory board.

☐ The authors and/or their immediate family members have management/advisory or consulting relationships noted in the "declaration of interests" section of the manuscript and on this form below.

Patents

We require that you disclose any patents related to this work by any of the authors or their institutions. ***Please select one of the following:***

☒ We, the authors and our immediate family members, have no related patents to declare.

☐ We, the authors, have a patent related to this work, which is noted in the "declaration of interests" section of the manuscript and on this form below, and we have noted the patents of immediate family members.

Please insert any "declaration of interests" statements in this space. This exact text should also be included in the "declaration of interests" section of the manuscript. If no authors have a competing interest, please insert the text, "The authors declare no competing interests."

Declaration of Interests
The authors declare no competing interests

Declaration of generative AI in scientific writing
During the preparation of this work the author(s) used ChatGPT for clarifying definitions of terms used across fields, as well as checking mathematical notation and coding. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the published article.

☒ **On behalf of all authors, I declare that I have disclosed all competing interests related to this work. If any exist, they have been included in the "declaration of interests" section of the manuscript.**